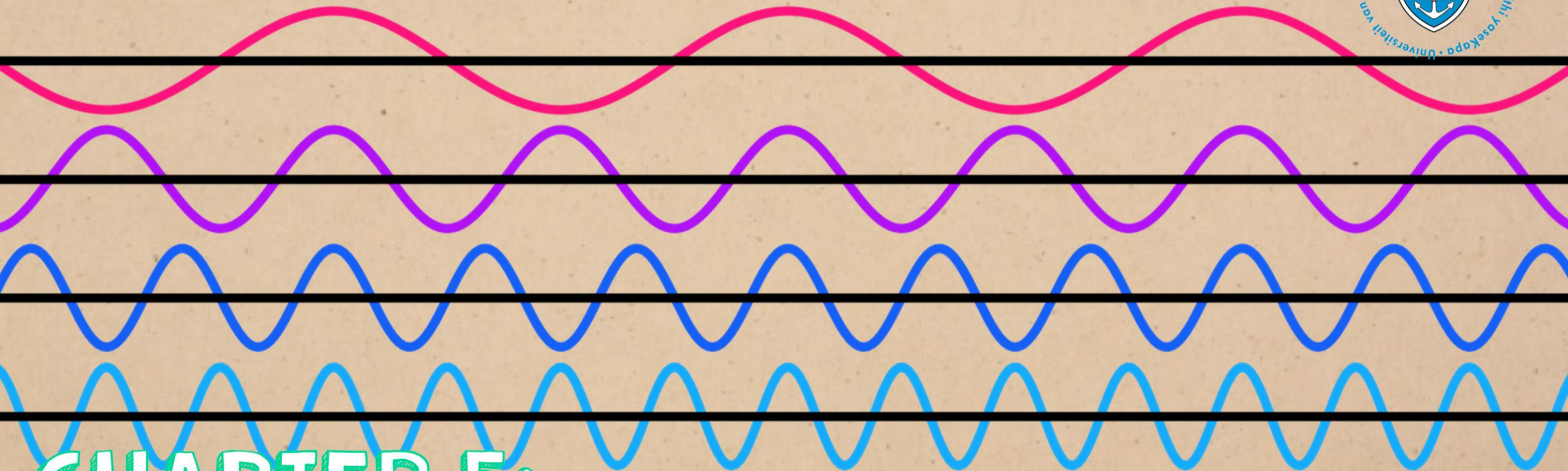




CHAPTER 5: FREQUENCY RESPONSE BODE PLOTS

RECAP

Our goal is to understand and manipulate how systems respond to inputs.



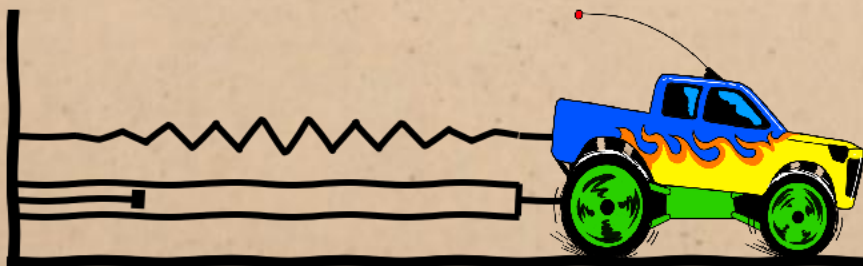
RECAP

There are **two ways** to model how linear and time-invariant (LTI) systems respond to inputs...



DYNAMIC RESPONSE

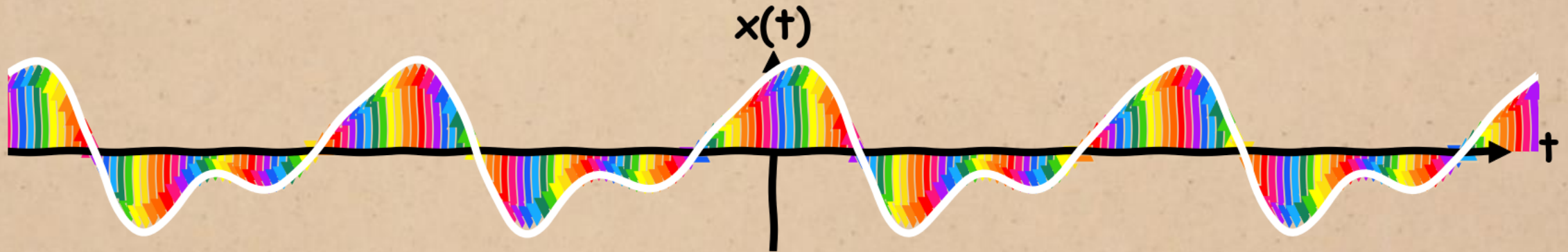
$$x(t) = e^{-t}u(t) - \frac{1}{2}e^{-2t}u(t) + \frac{1}{2}$$



One is to consider the system's
time domain response.

BASIS

We think of time-domain signals as **weighted sums** of scaled, shifted impulses.

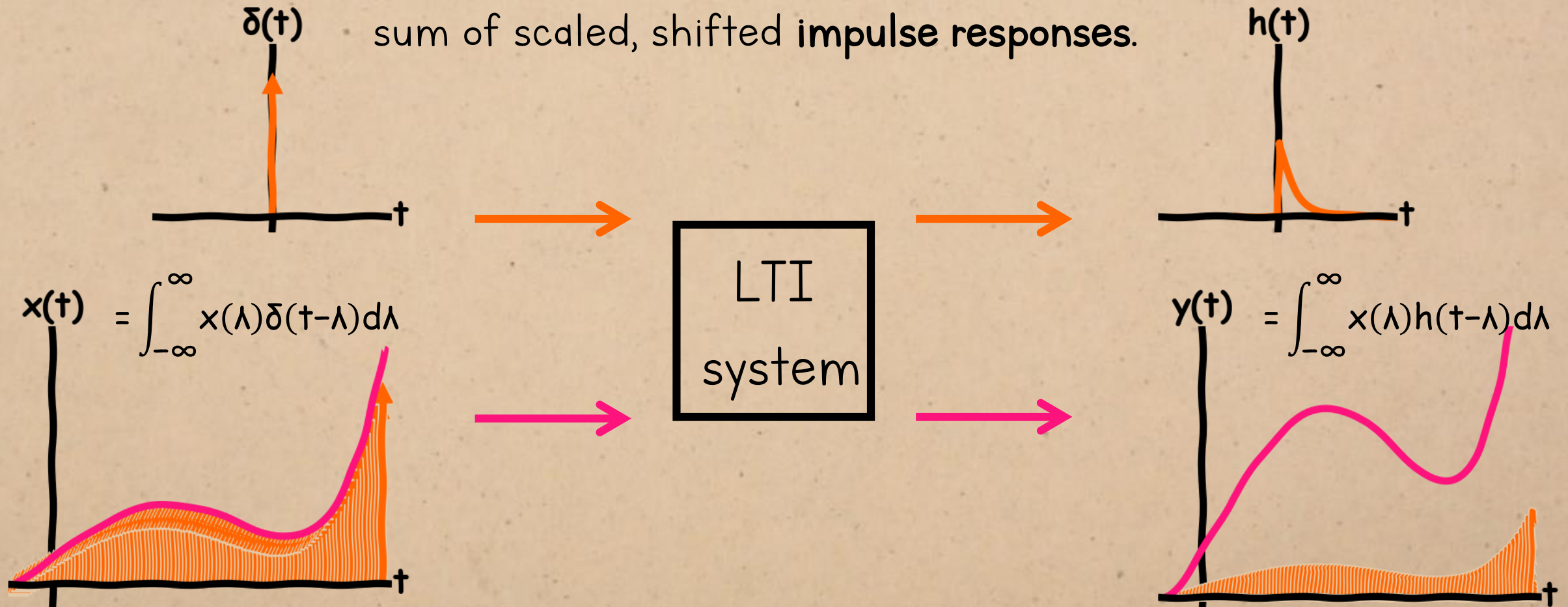


$$= \dots x(-1) \delta(t+1) + \dots x(0) \delta(t) + \dots x(1) \delta(t-1) + \dots x(2) \delta(t-2) + \dots x(3) \delta(t-3) + \dots$$

$$= \int_{-\infty}^{\infty} x(\lambda) \delta(t-\lambda) d\lambda$$

IMPULSE

...and therefore, the response is a weighted sum of scaled, shifted impulse responses.



RECAP

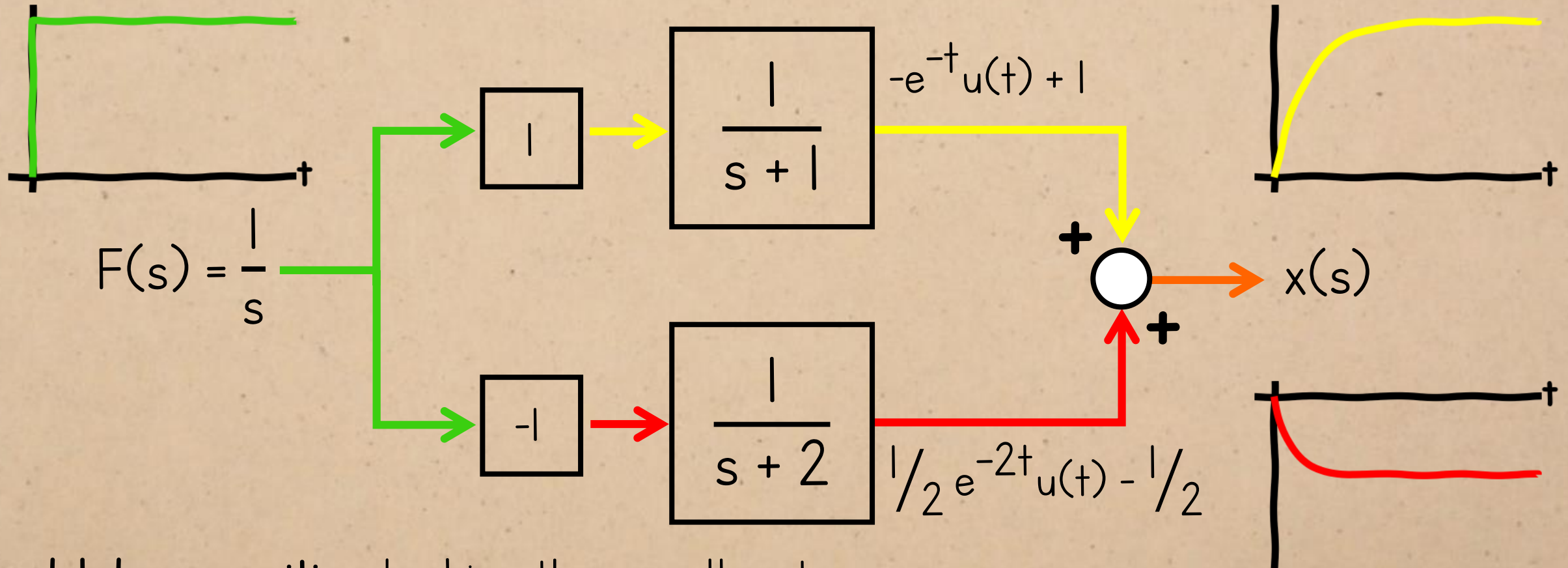
Our analysis so far has largely focussed on the **poles and zeros** of the system.

zeros = roots of the numerator

$$\frac{(s - z_1)(s - z_2) \dots (s - z_m)}{(s - p_1)(s - p_2) \dots (s - p_n)}$$

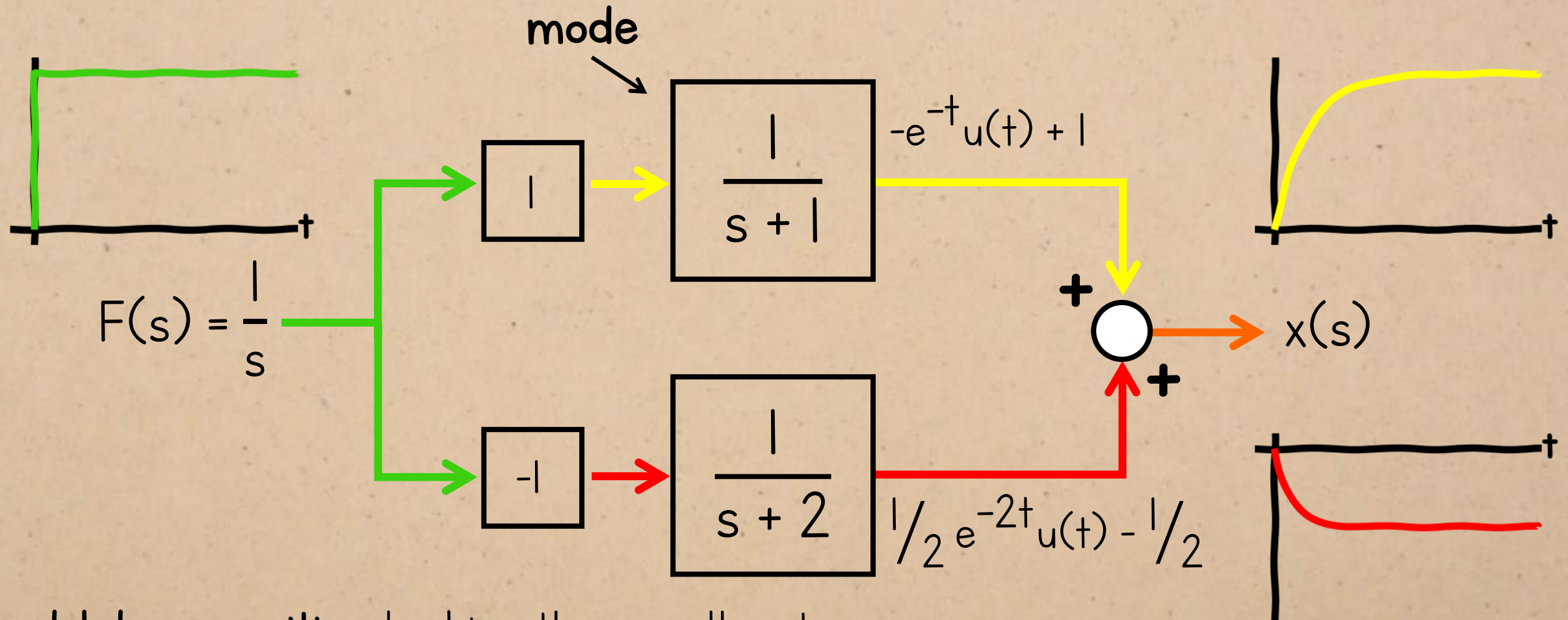
poles = roots of the denominator

DYNAMIC RESPONSE



modal decomposition: breaking the overall system response into the responses to individual poles or pole-pairs

DYNAMIC RESPONSE



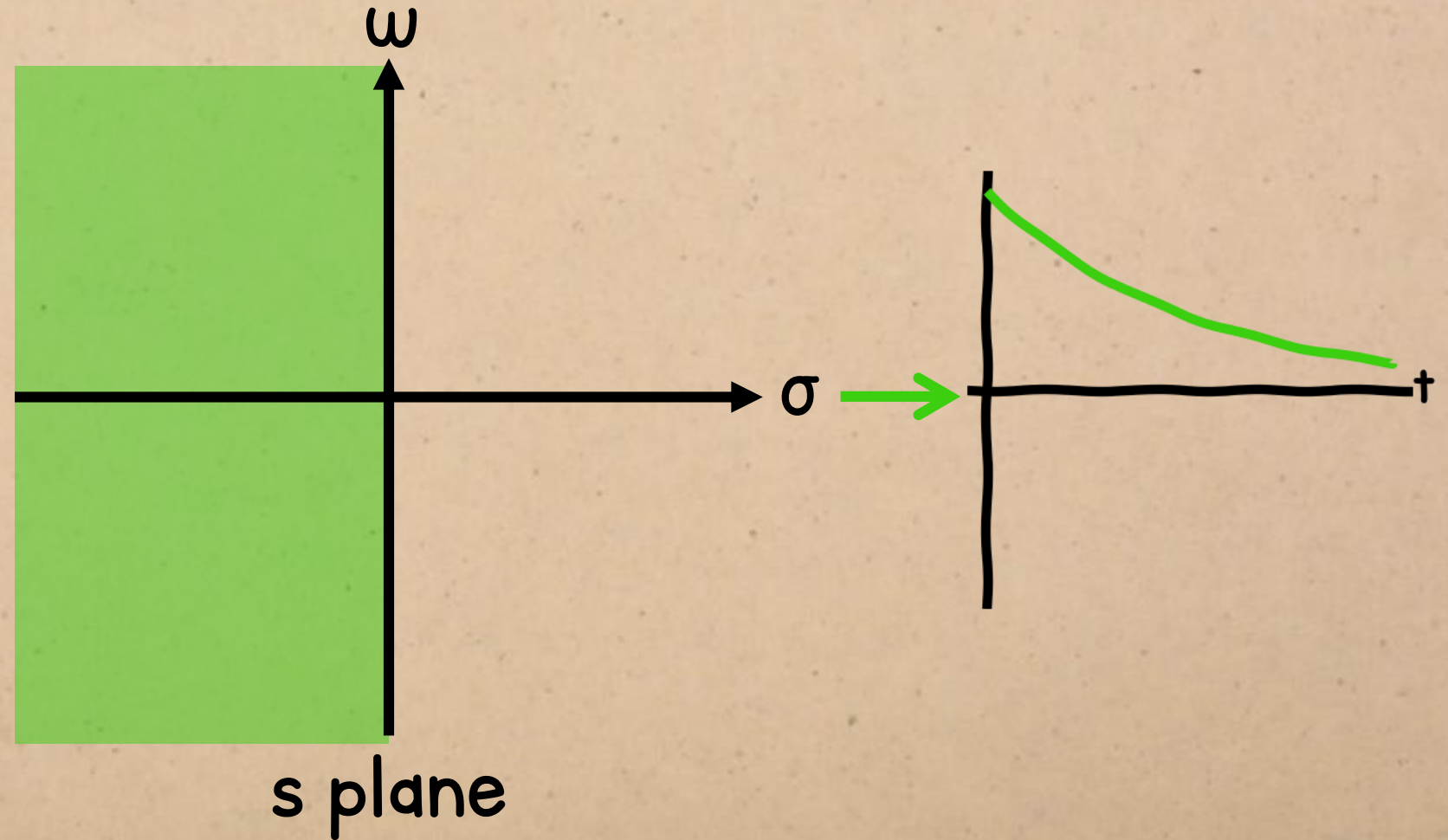
modal decomposition: breaking the overall system response into the responses to individual poles or pole-pairs

The location of the pole in the s -plane determines the characteristics of its impulse response.

For example...

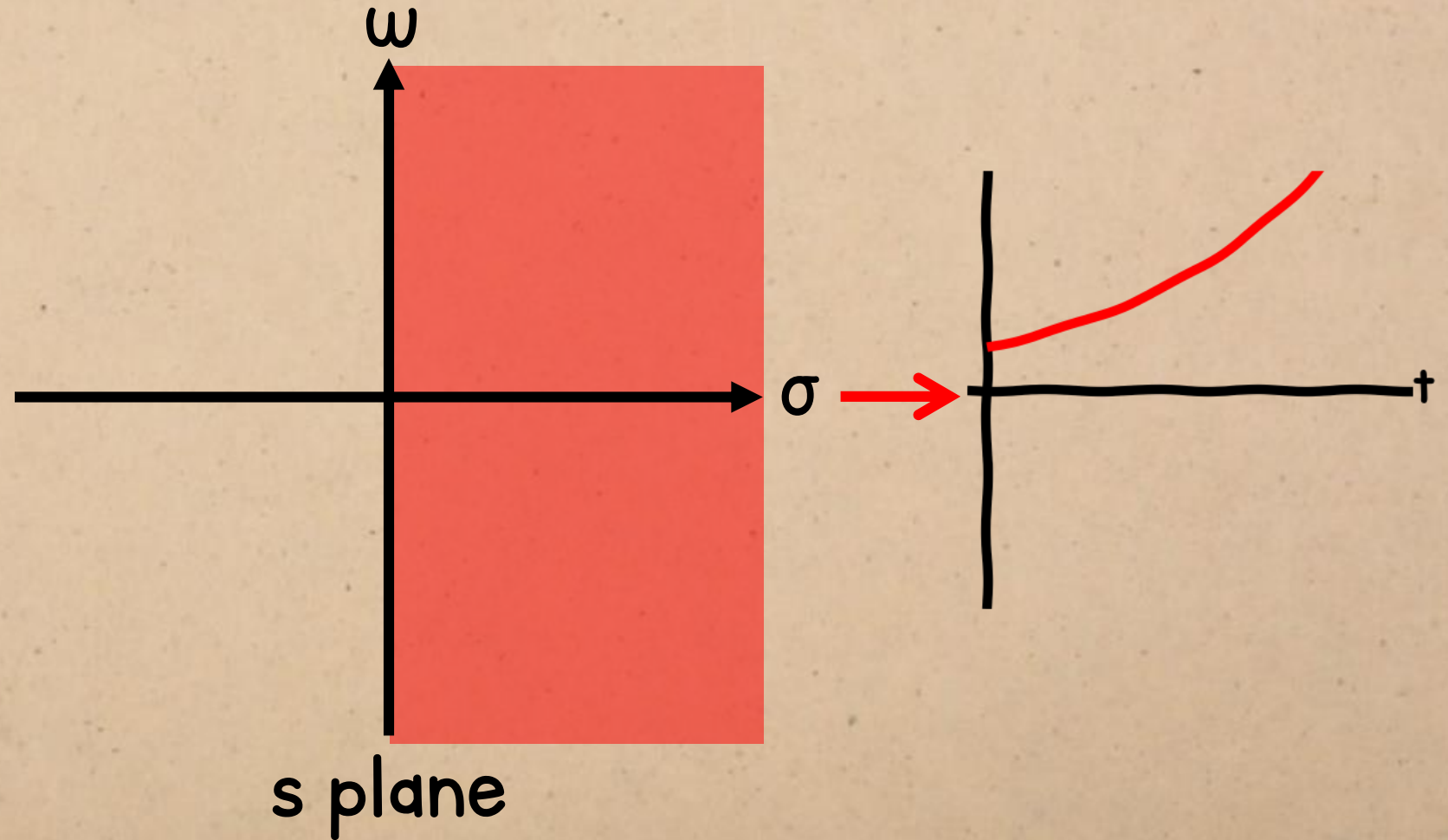
TRANSIENT RESPONSE

Poles in the **left-half plane** (negative poles) are **stable**.



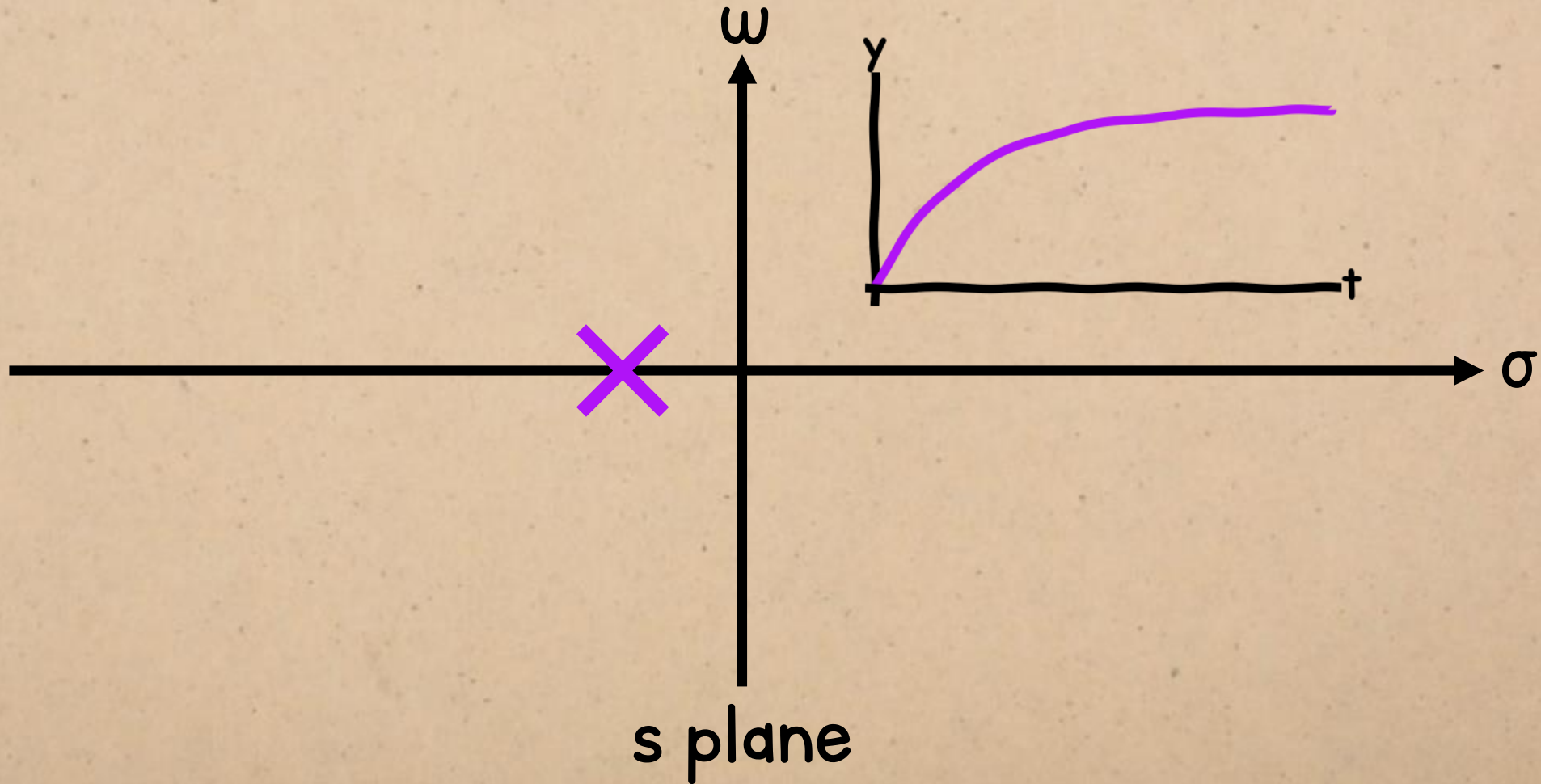
TRANSIENT RESPONSE

Poles in the right-half plane (positive poles) are unstable.



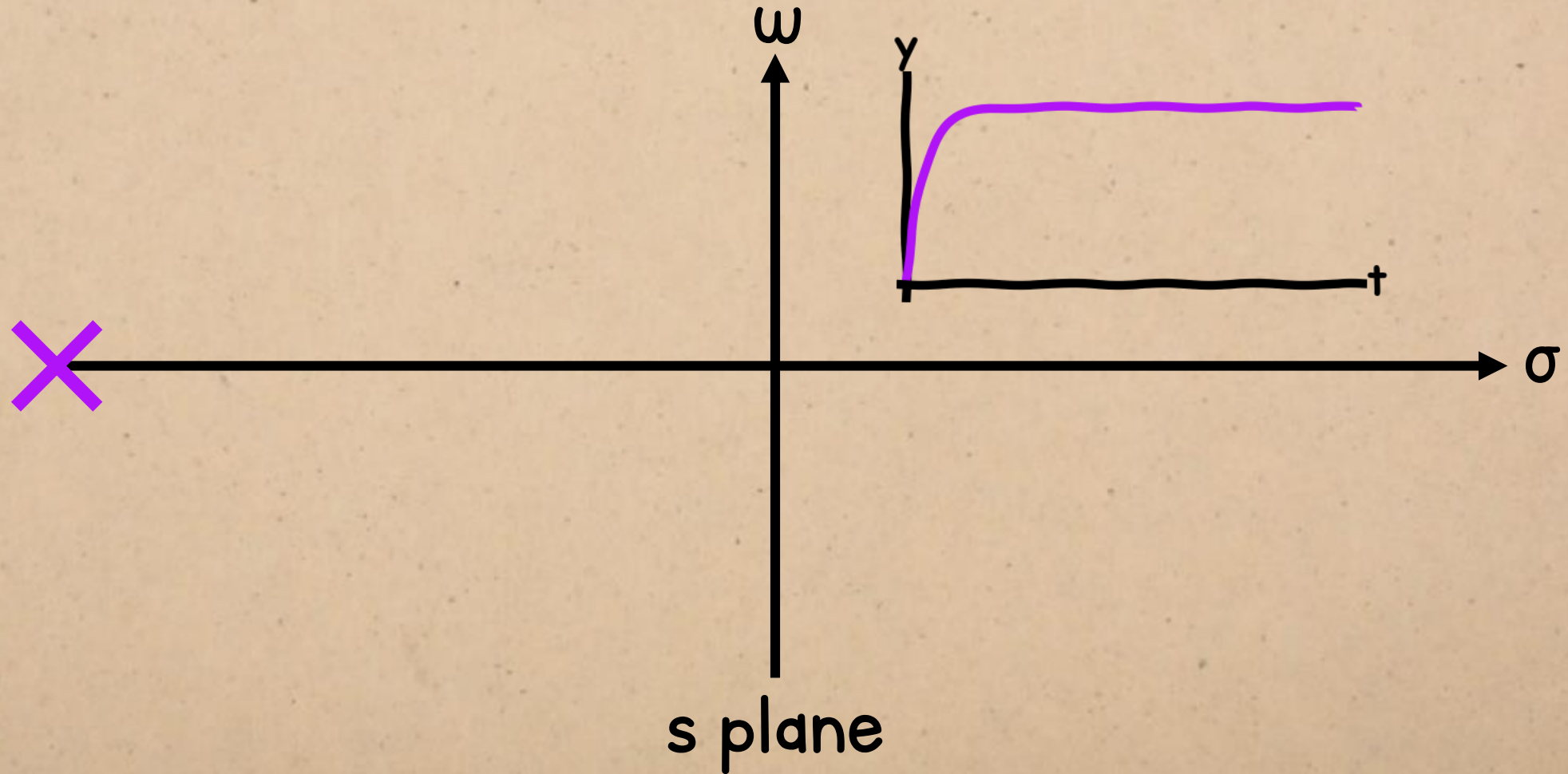
FIRST-ORDER SYSTEM

The distance from the pole to the origin determines the time constant.



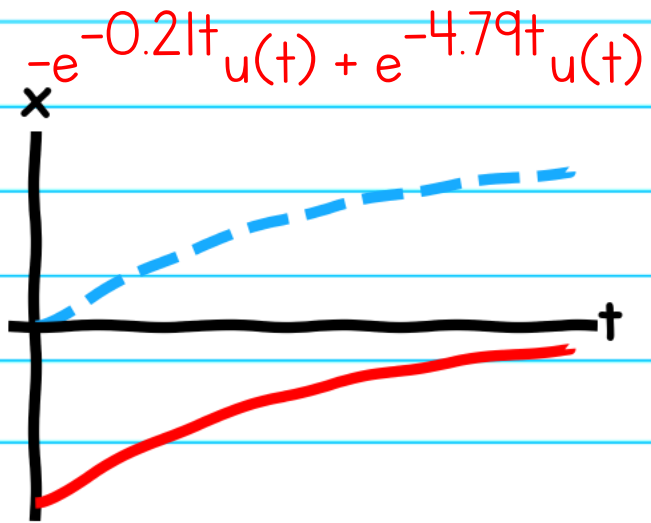
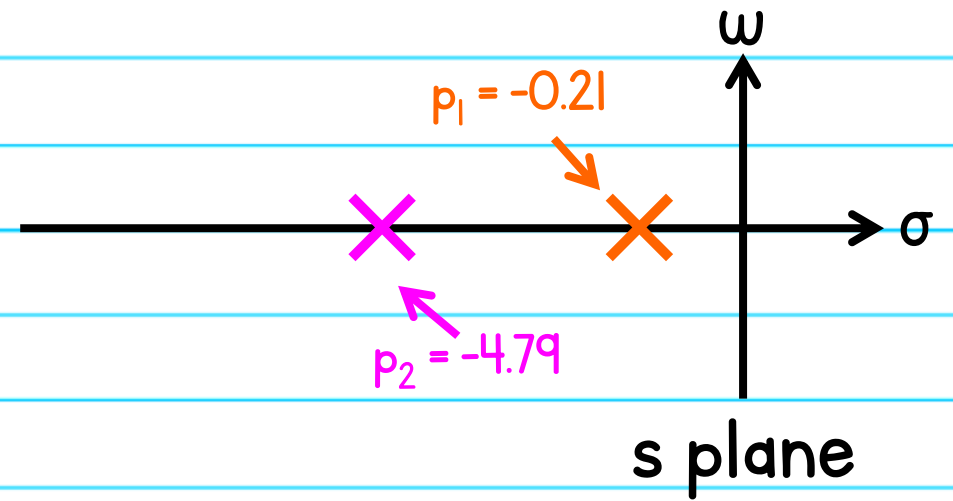
FIRST-ORDER SYSTEM

The distance from the pole to the origin determines the time constant.



FORM OF RESPONSE

★ $b^2 - 4mk > 0 \rightarrow$ two real poles



$$b^2 - 4 > 0$$



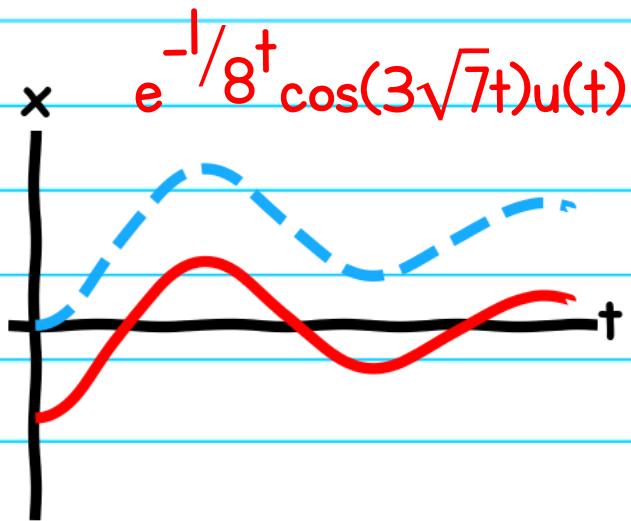
FORM OF RESPONSE

- ★ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ★ $b = 2m\omega_n \rightarrow$ repeated real pole
- ★ $b^2 - 4mk < 0 \rightarrow$ complex pole pair

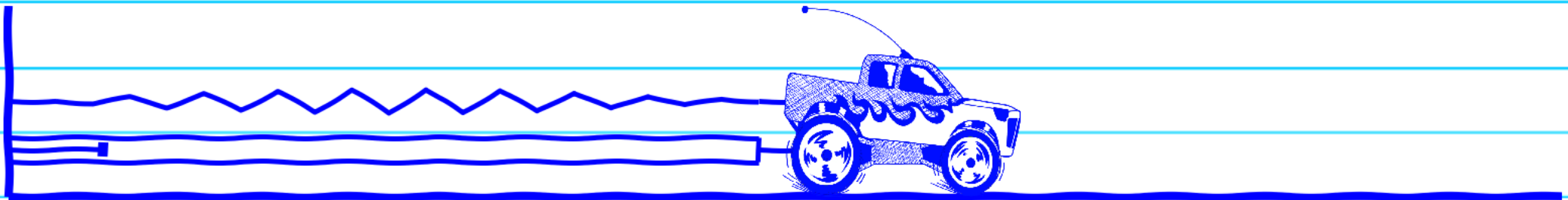
$$p_1 = -1/8 + j3\sqrt{7}$$

$$p_2 = -1/8 - j3\sqrt{7}$$

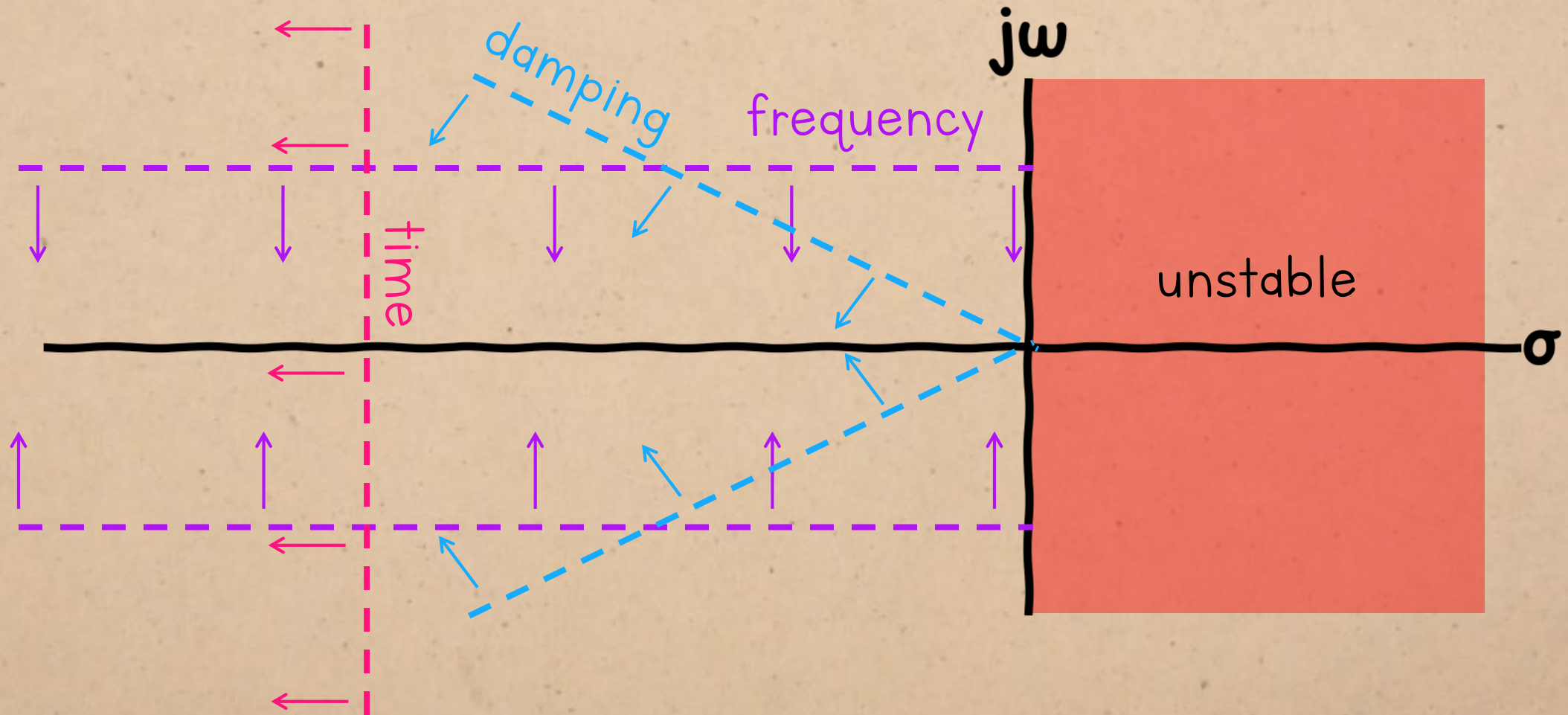
s plane



$$b^2 - 4 < 0$$



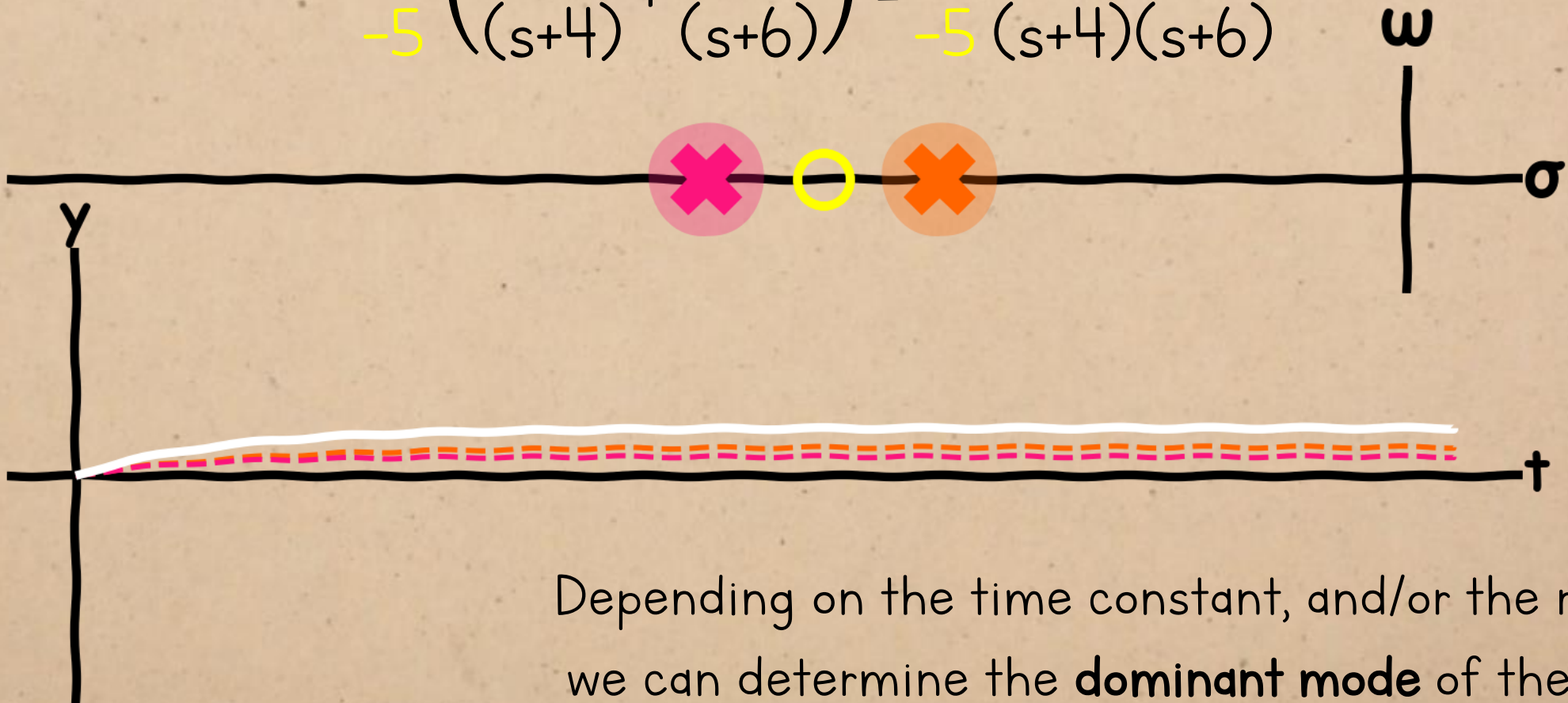
DESIGN GUIDE TO THE S-PLANE



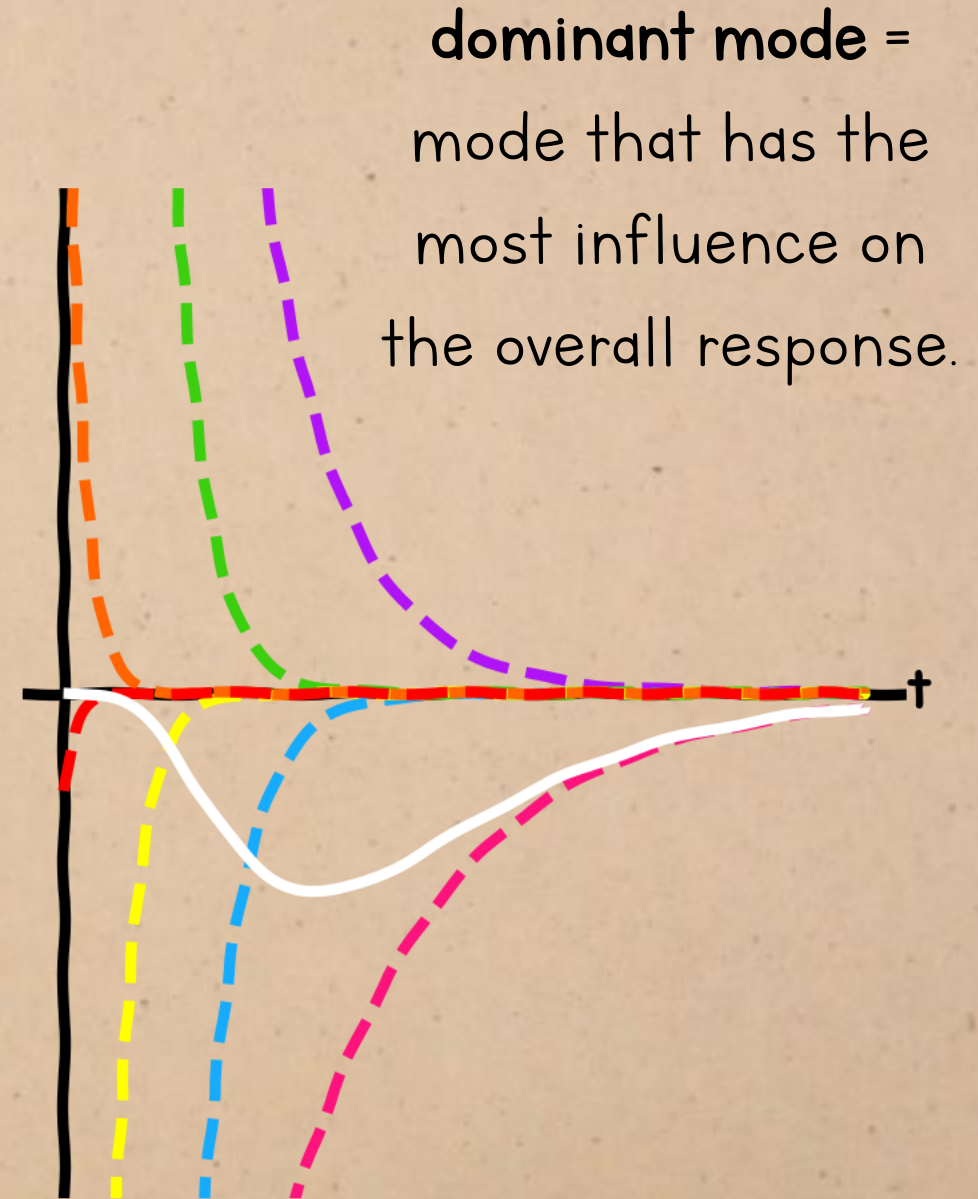
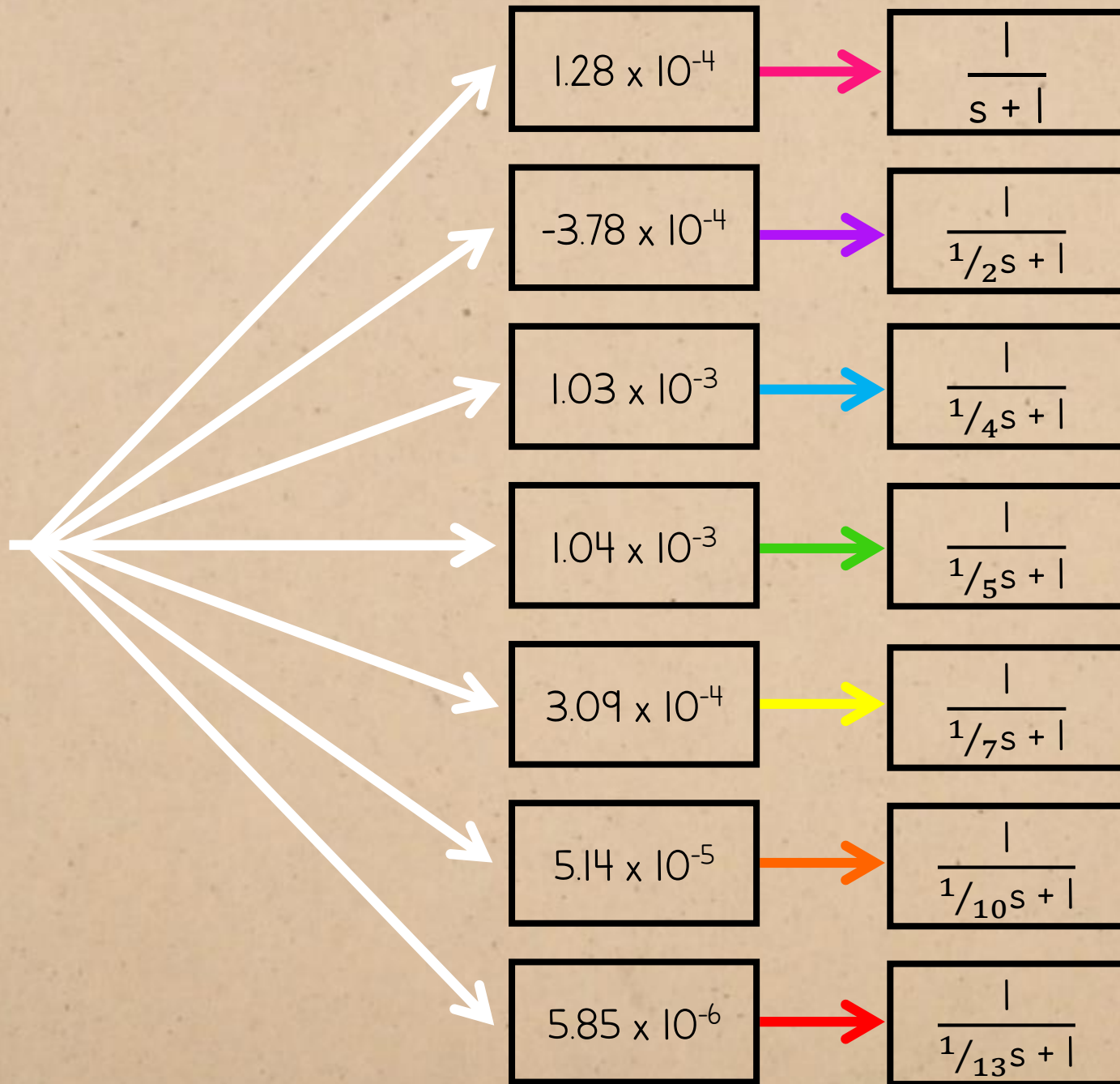
We also examined
how the zeros affect
the overall response...

ZEROS

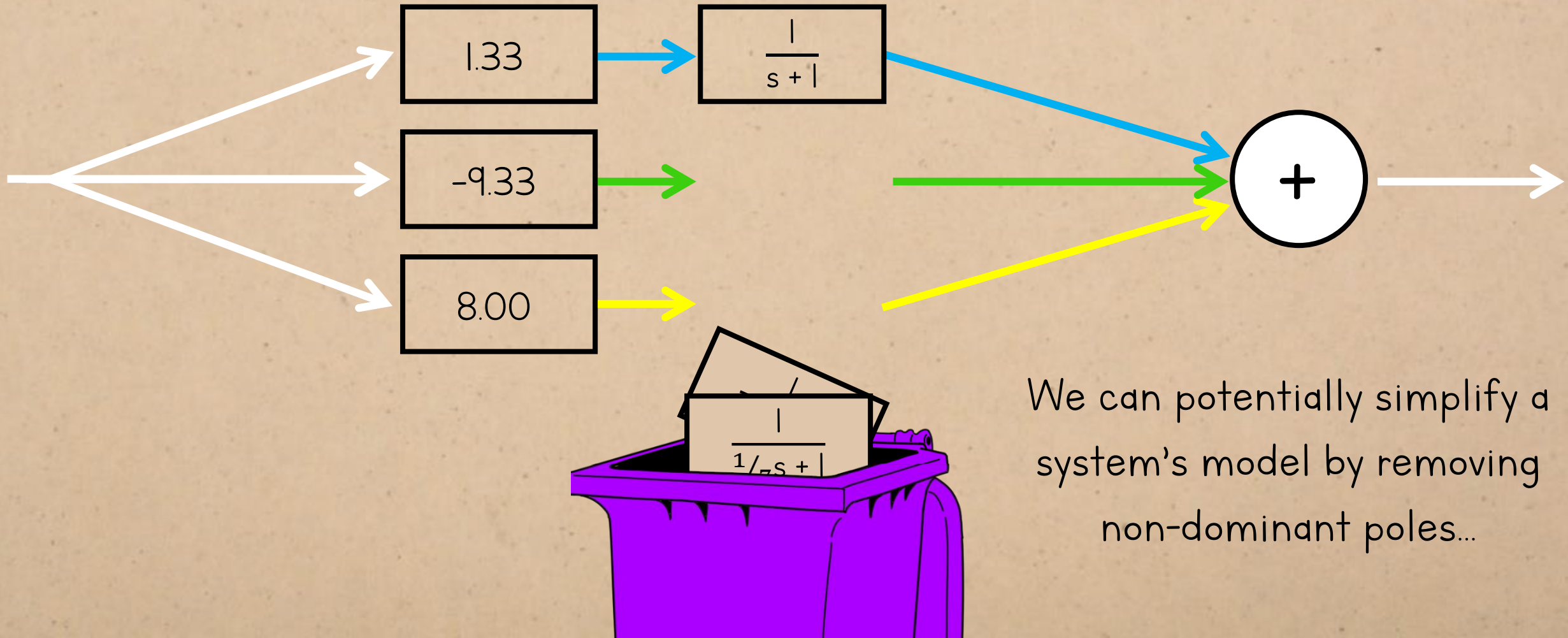
$$\frac{1}{-5} \left(\frac{\frac{1}{2}}{(s+4)} + \frac{\frac{1}{2}}{(s+6)} \right) = \frac{1}{-5} \frac{(s+5)}{(s+4)(s+6)}$$



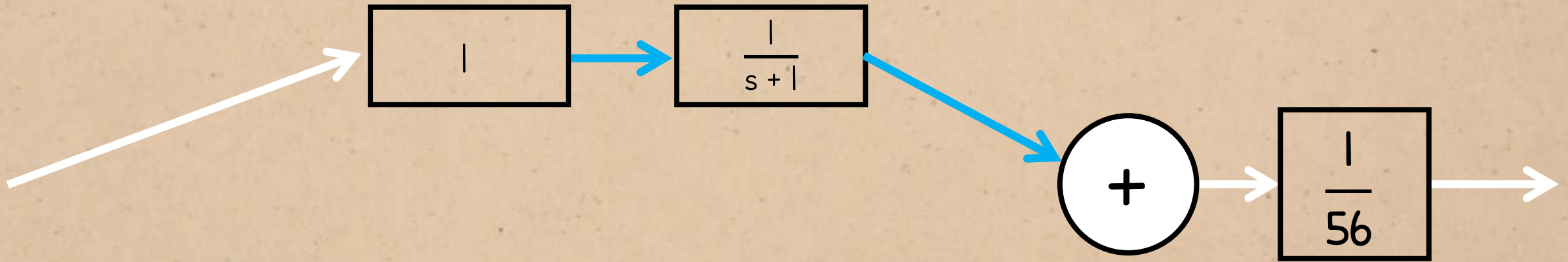
Depending on the time constant, and/or the residuals,
we can determine the **dominant mode** of the system.



ORDER REDUCTION

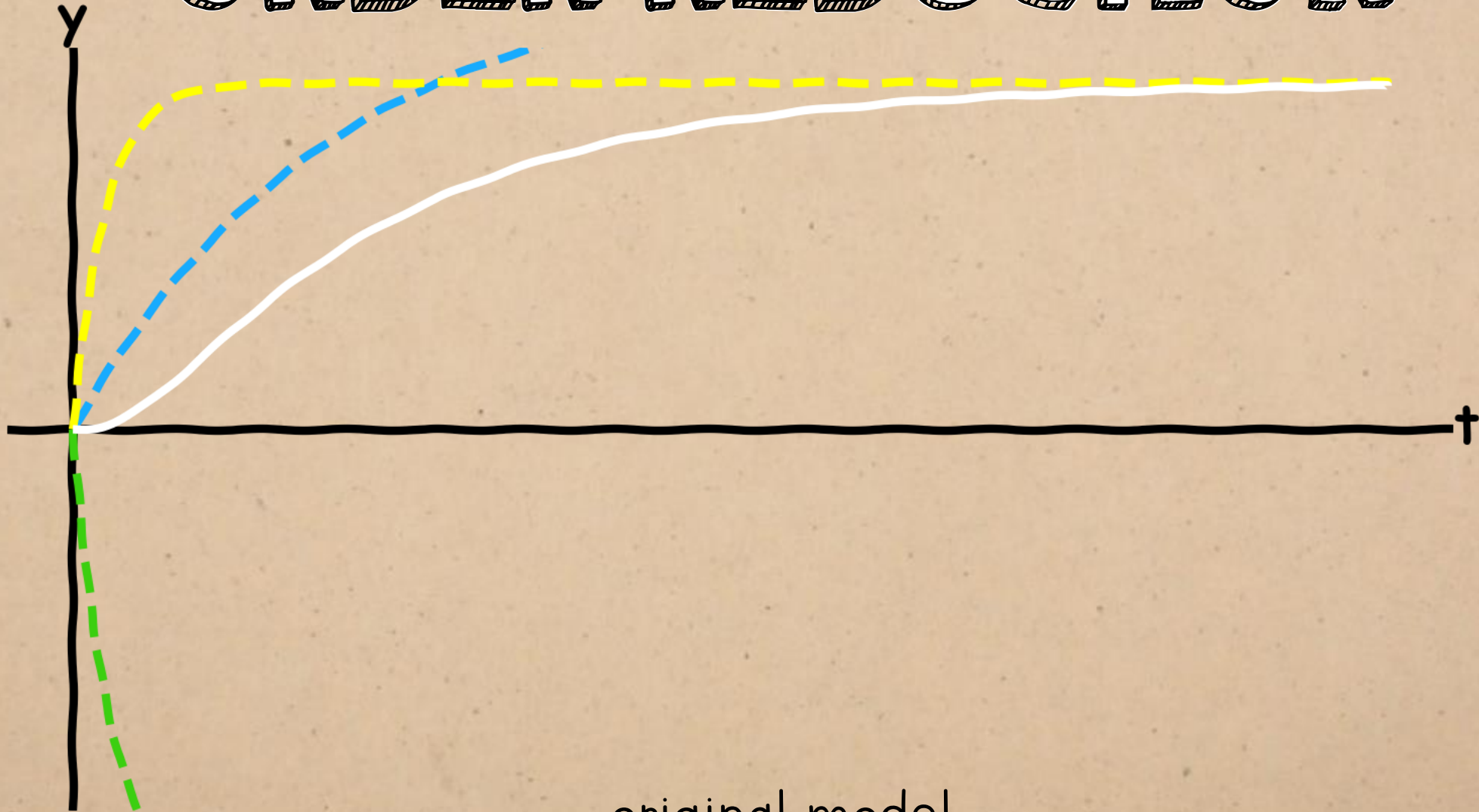


ORDER REDUCTION



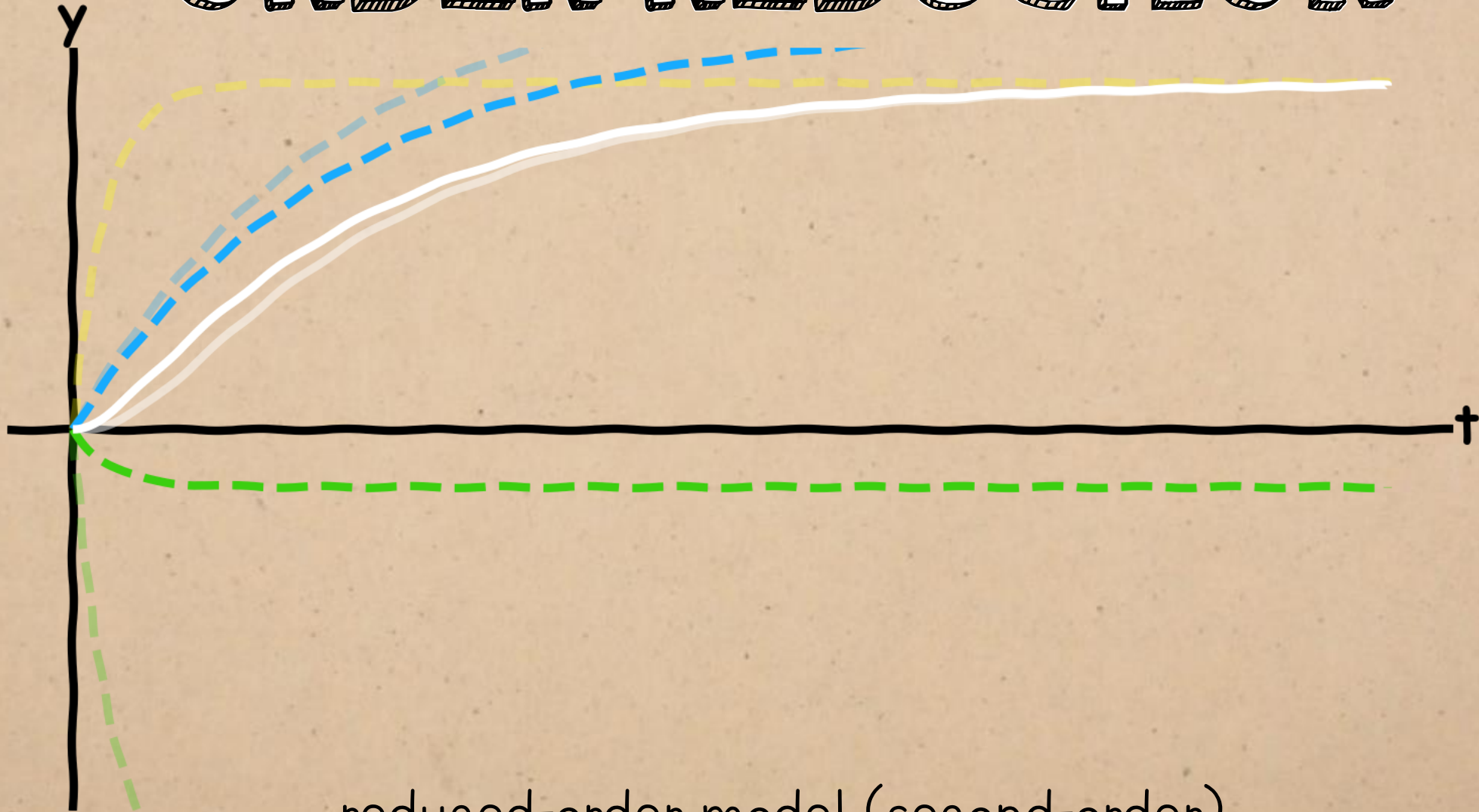
...and replacing them with the **gains** of the pole terms.

ORDER REDUCTION



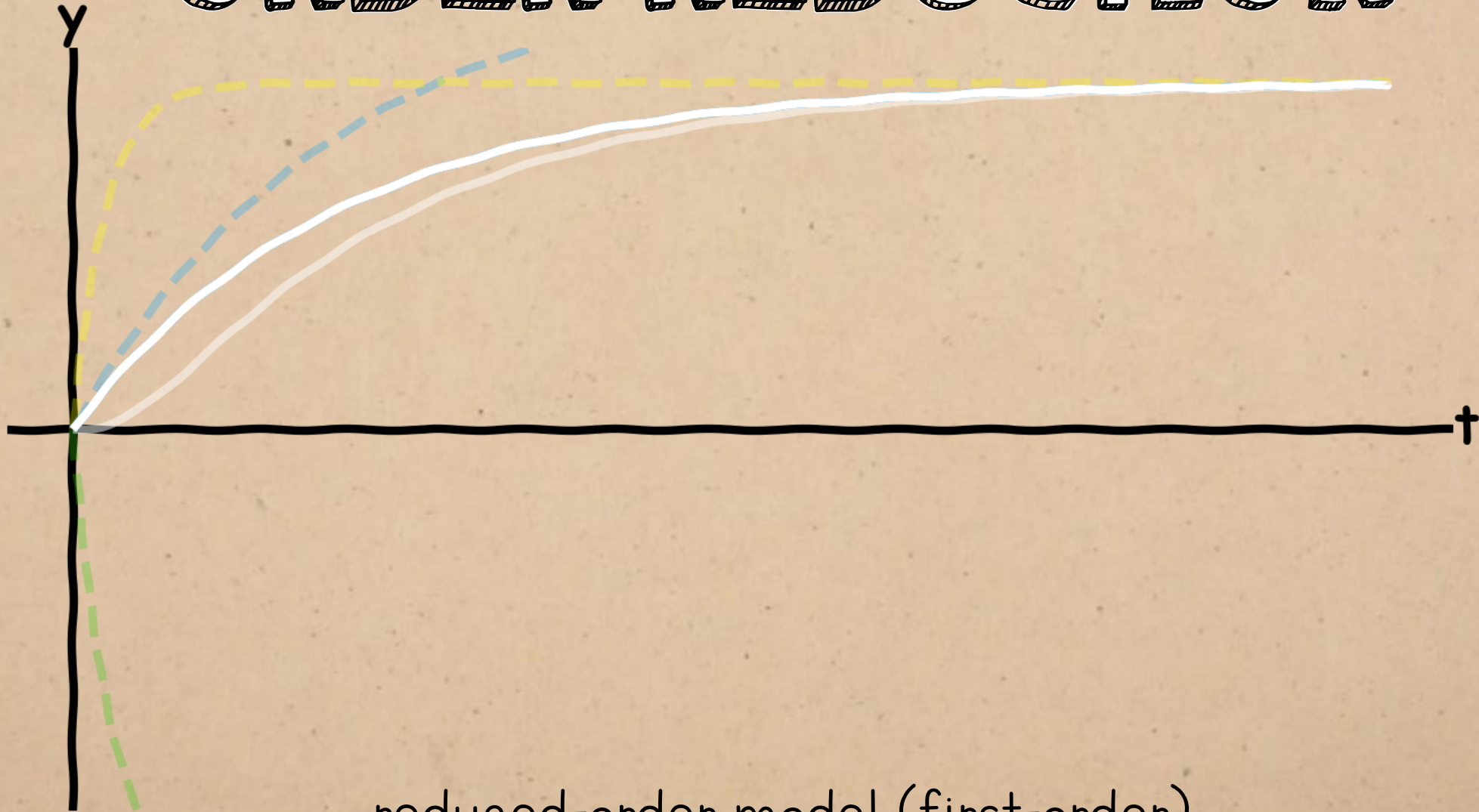
original model

ORDER REDUCTION



reduced-order model (second-order)

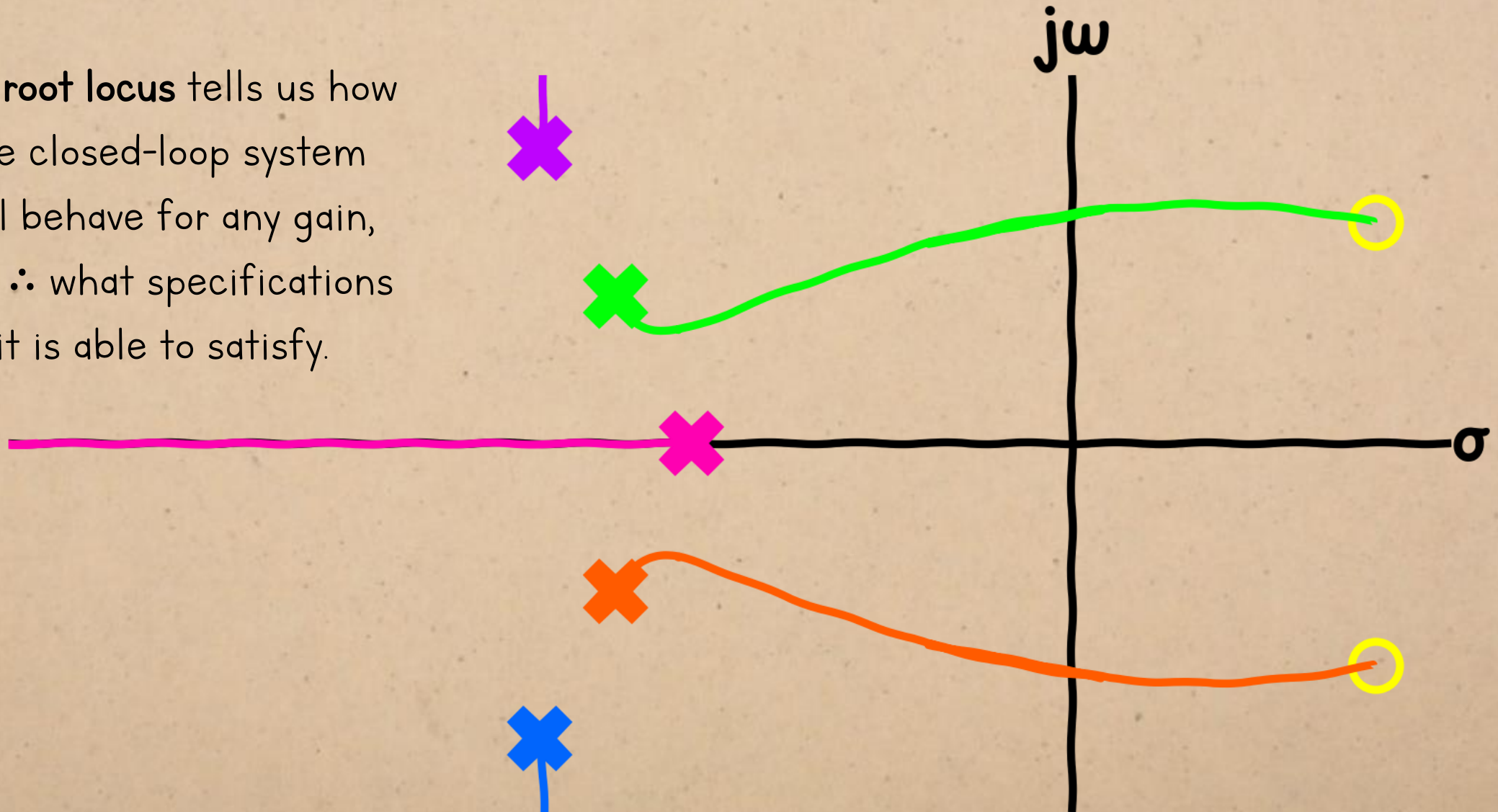
ORDER REDUCTION



reduced-order model (first-order)

RECAP

The **root locus** tells us how the closed-loop system will behave for any gain, and $\therefore$ what specifications it is able to satisfy.

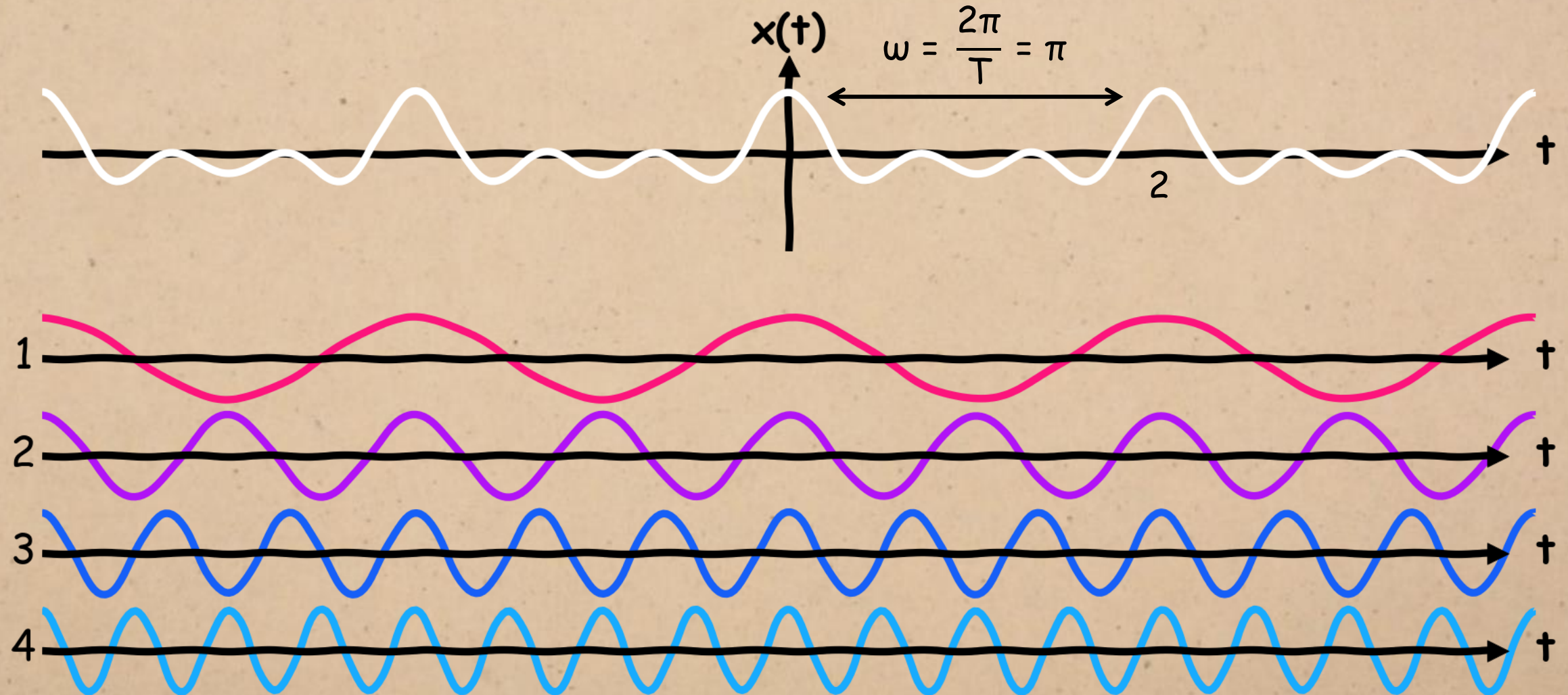


TL;DR

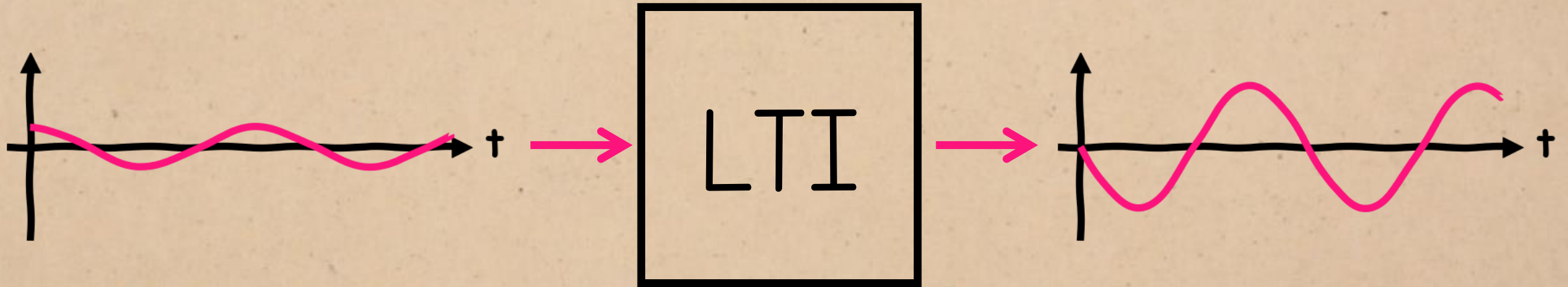
Pole-zero analysis methods can be considered
time-domain techniques.

(because they focus on the time-domain impulse response)

We can also model signals as a weighted sum of **frequency components**
(complex exponentials, or sinusoids for real signals.)

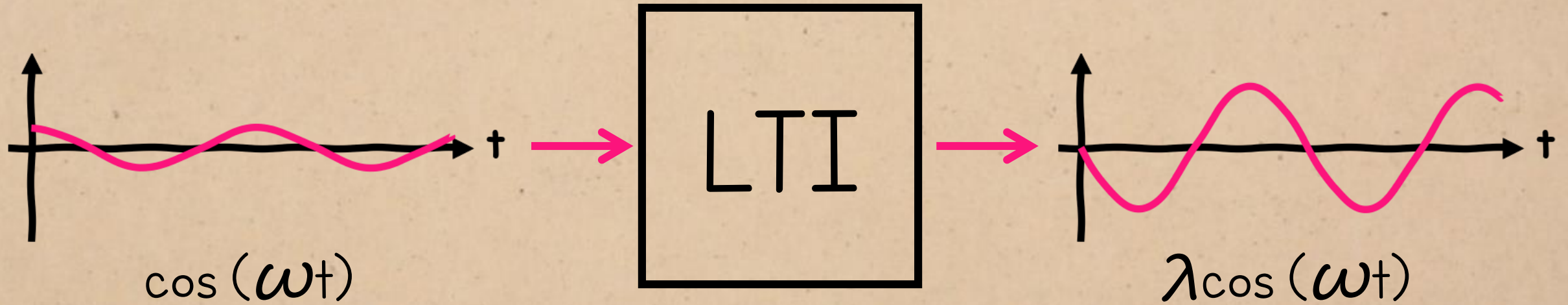


Sinusoids are **eigenfunctions** of LTI systems.

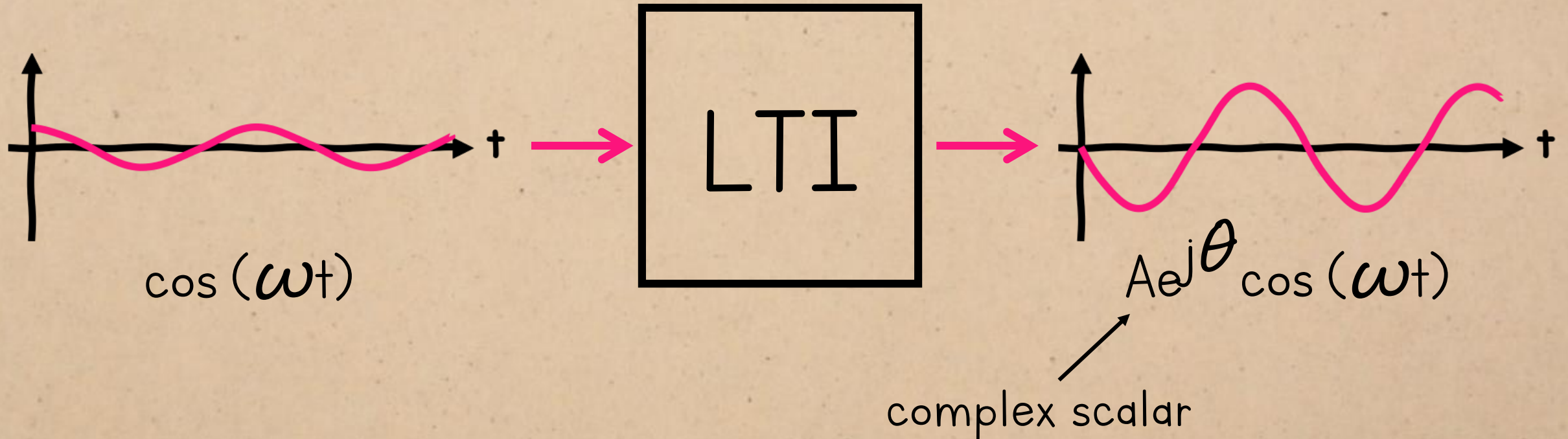


(They can pass through without changing shape or frequency)

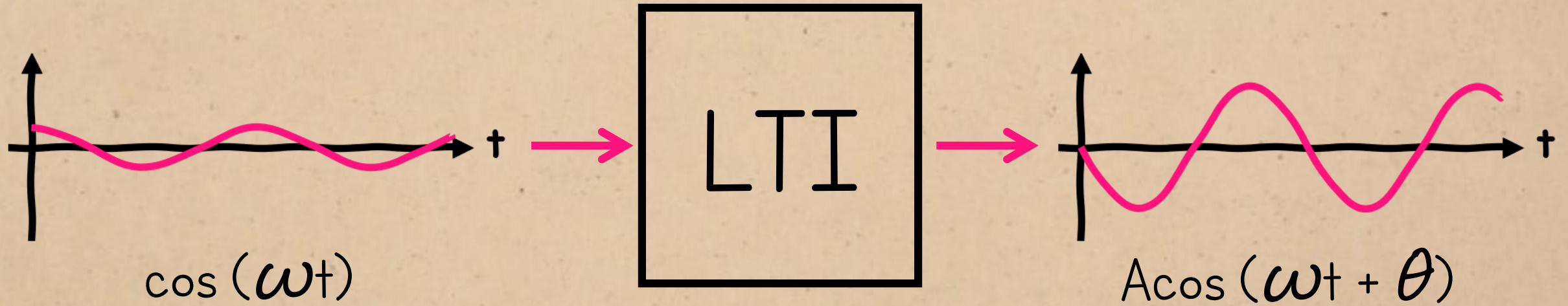
The response to a sinusoid will be the same sinusoid multiplied by a scalar (eigenvalue).



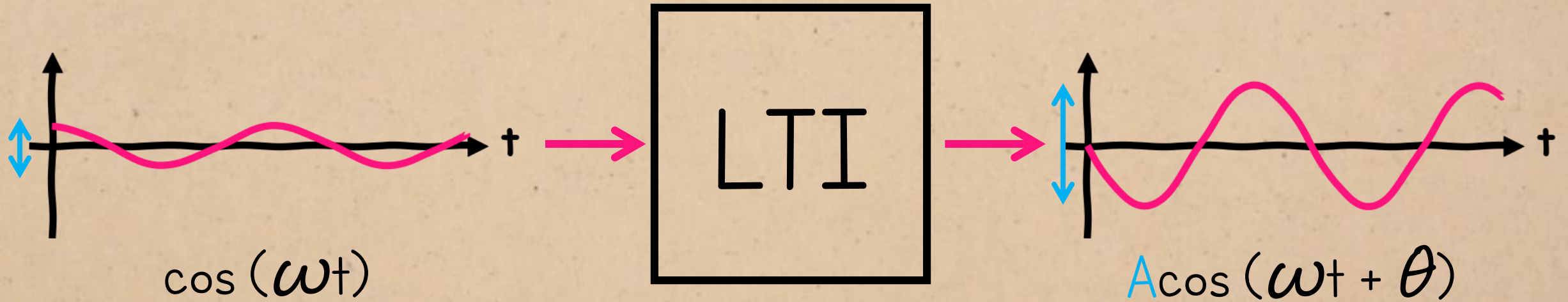
The response to a sinusoid will be the same sinusoid multiplied by a scalar (eigenvalue).



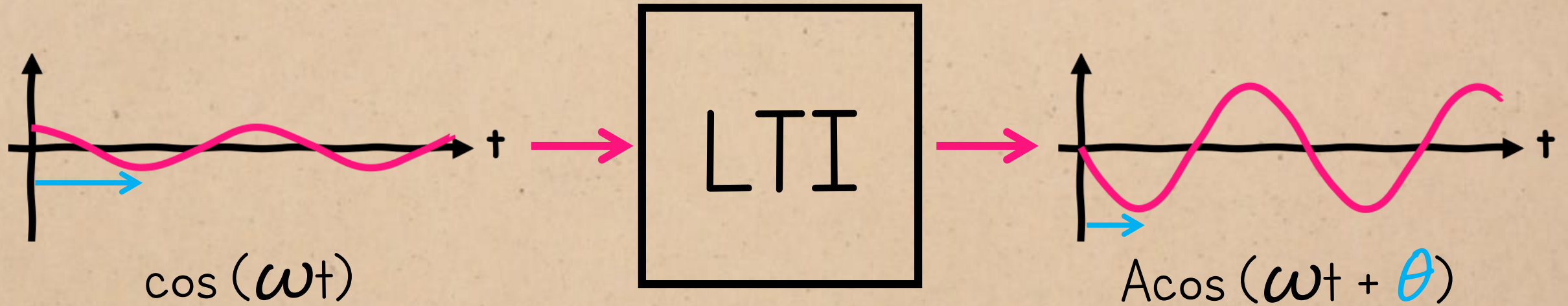
$\therefore$ the system can change the **magnitude** and **phase** of the sinusoid.



∴ the system can change the **magnitude** and **phase** of the sinusoid.

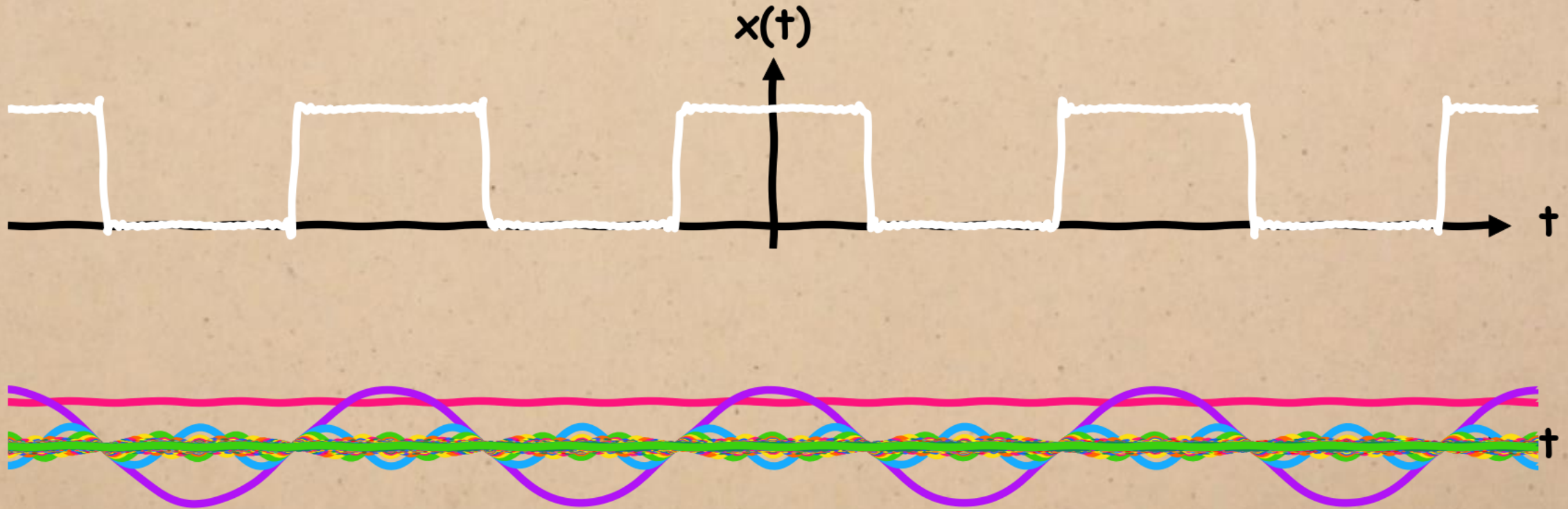


∴ the system can change the **magnitude** and **phase** of the sinusoid.



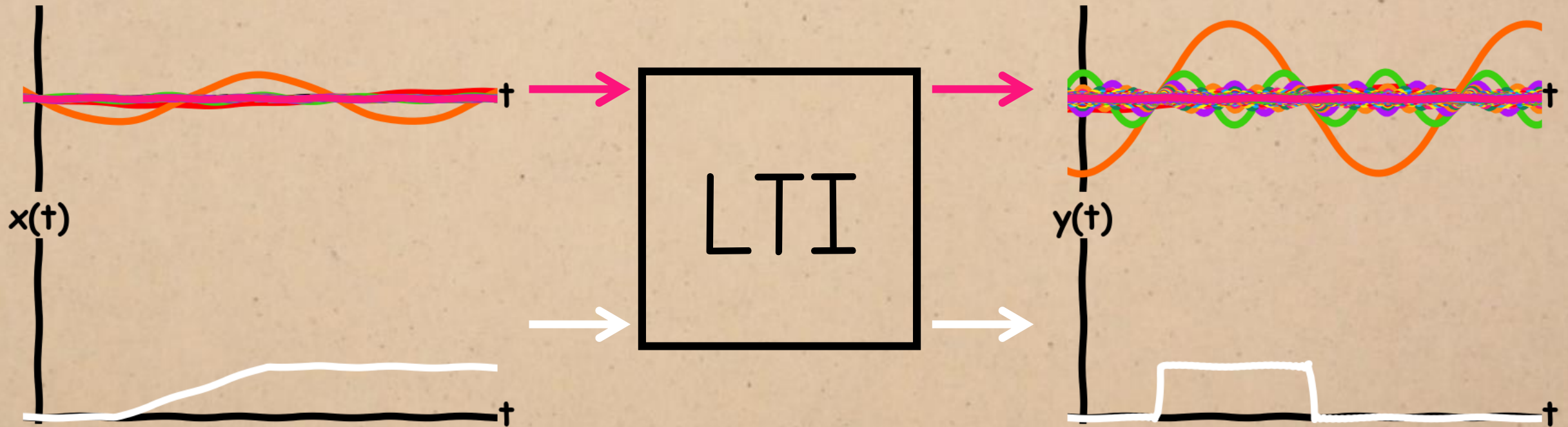
FOURIER SERIES

Because we can form any signal from these
frequency components...



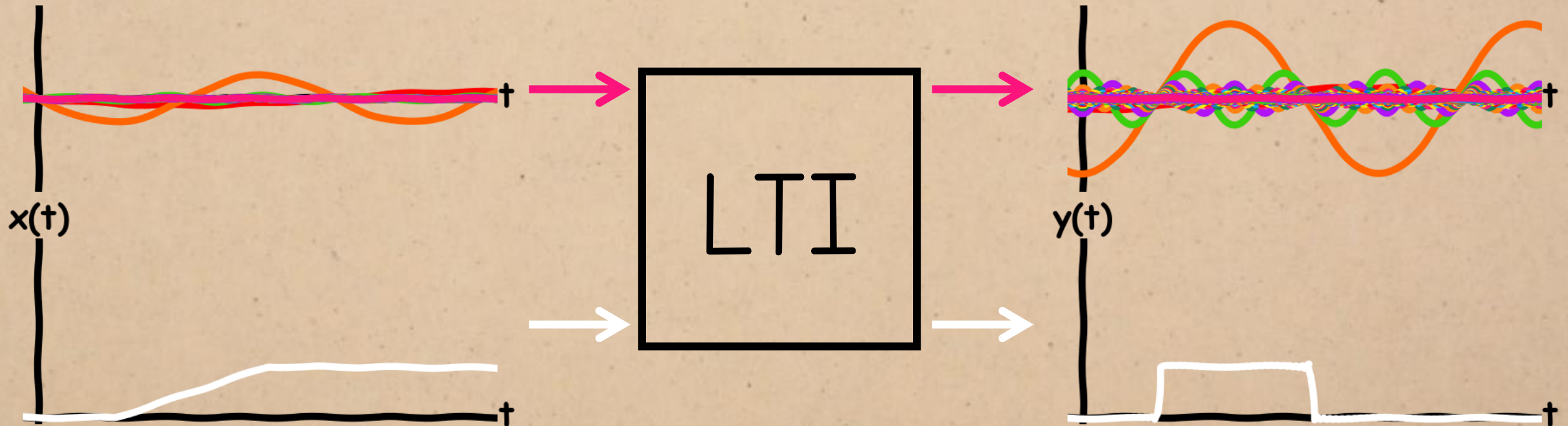
60 harmonics

...If we know how the system affects the magnitude and phase of a sinusoid at any frequency, we can completely define its response.



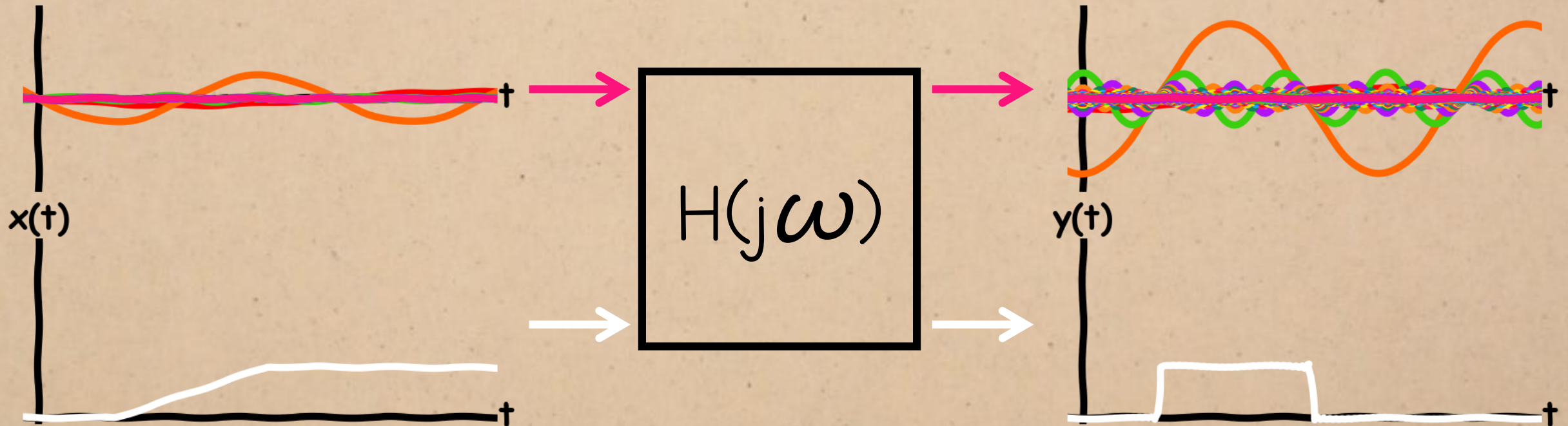
FREQUENCY RESPONSE

how the system affects the magnitude and phase of a sinusoid at any frequency



FREQUENCY RESPONSE

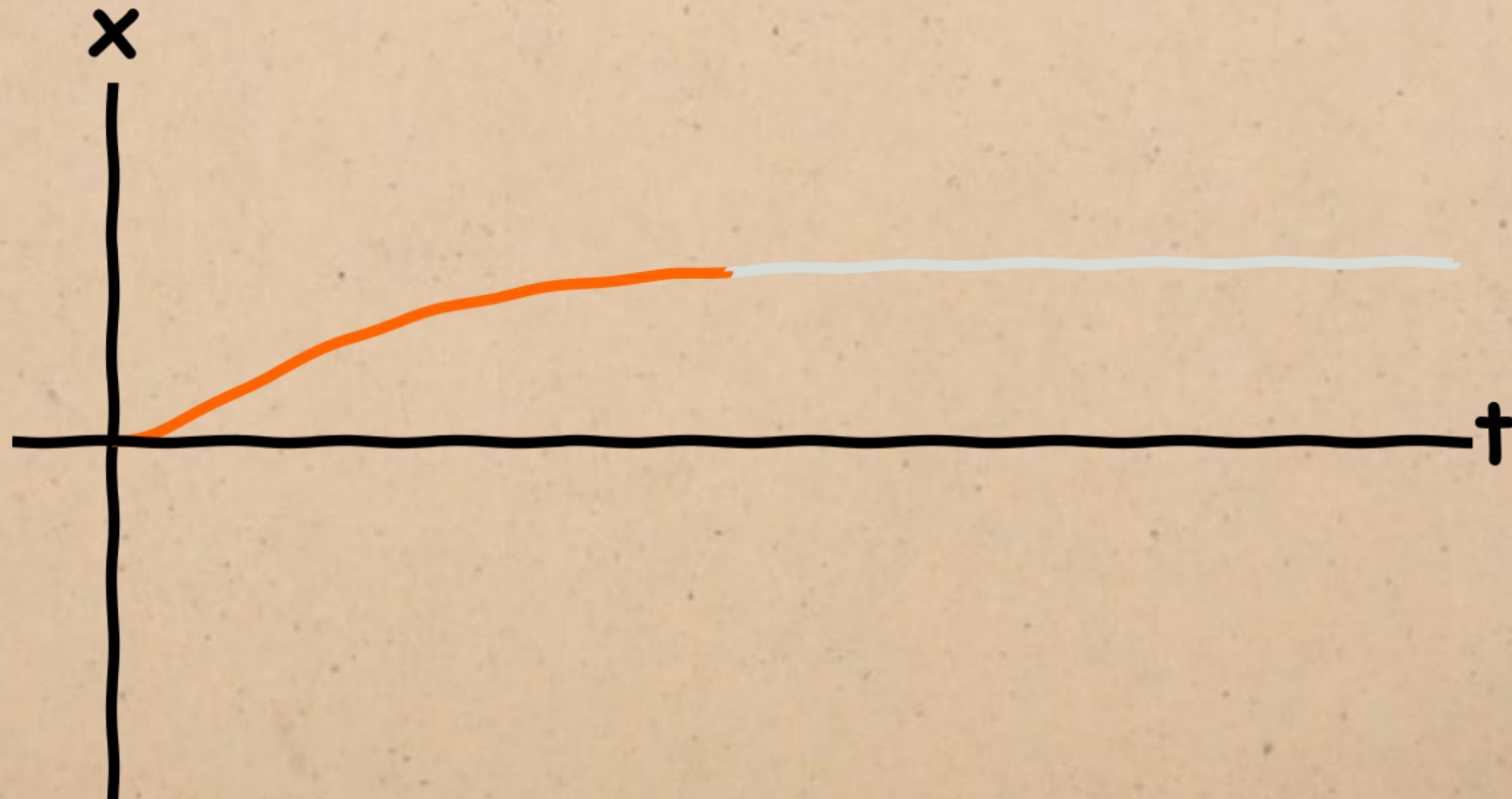
We can find an expression for the frequency response by substituting $s = j\omega$ into the transfer function.



We could break the time-domain response into different parts...

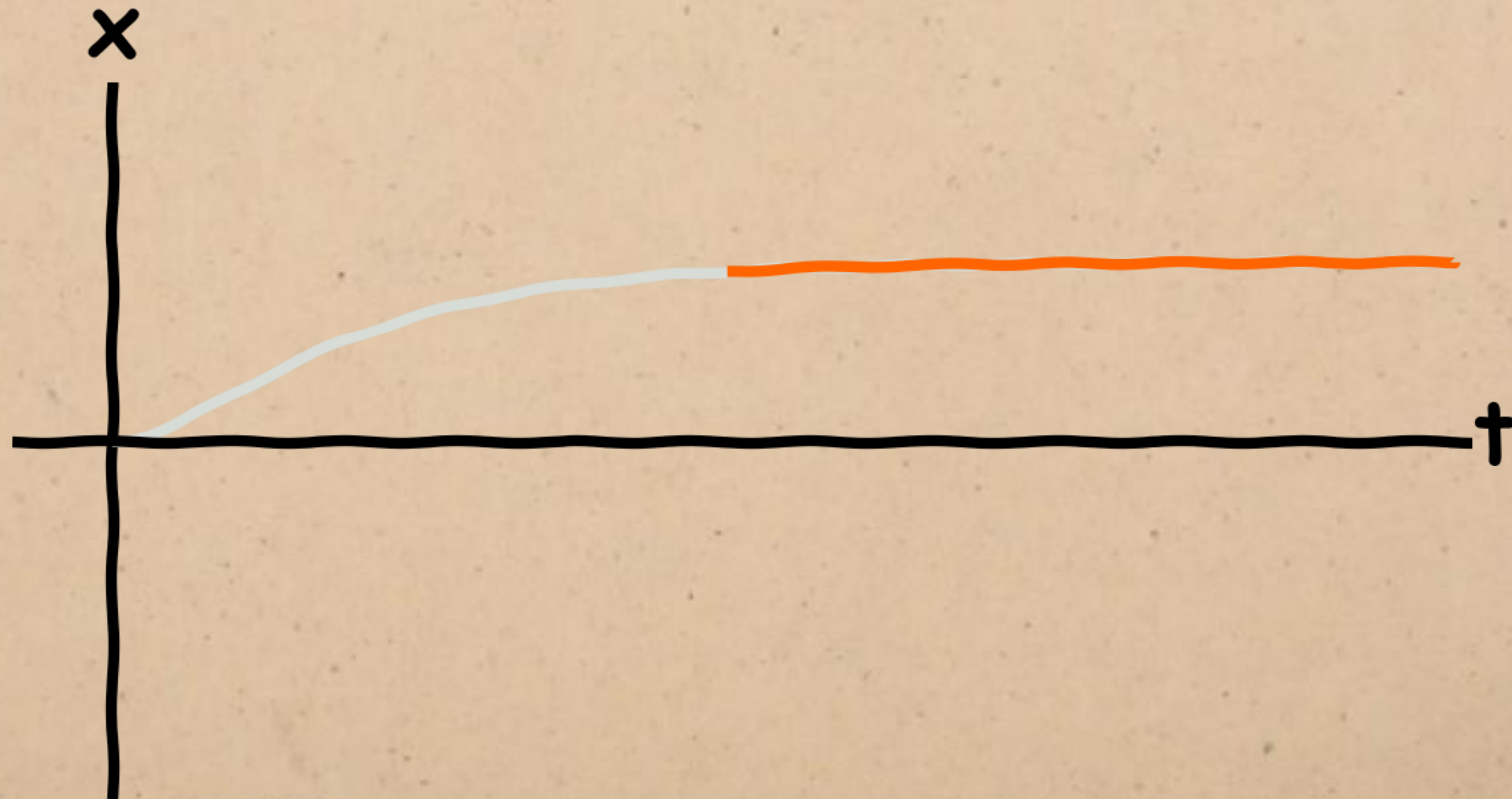
TRANSIENT RESPONSE

transient response = part of the response that decays away over time



STEADY-STATE RESPONSE

steady state = part of the response that remains as time tends to infinity.



DYNAMIC RESPONSE

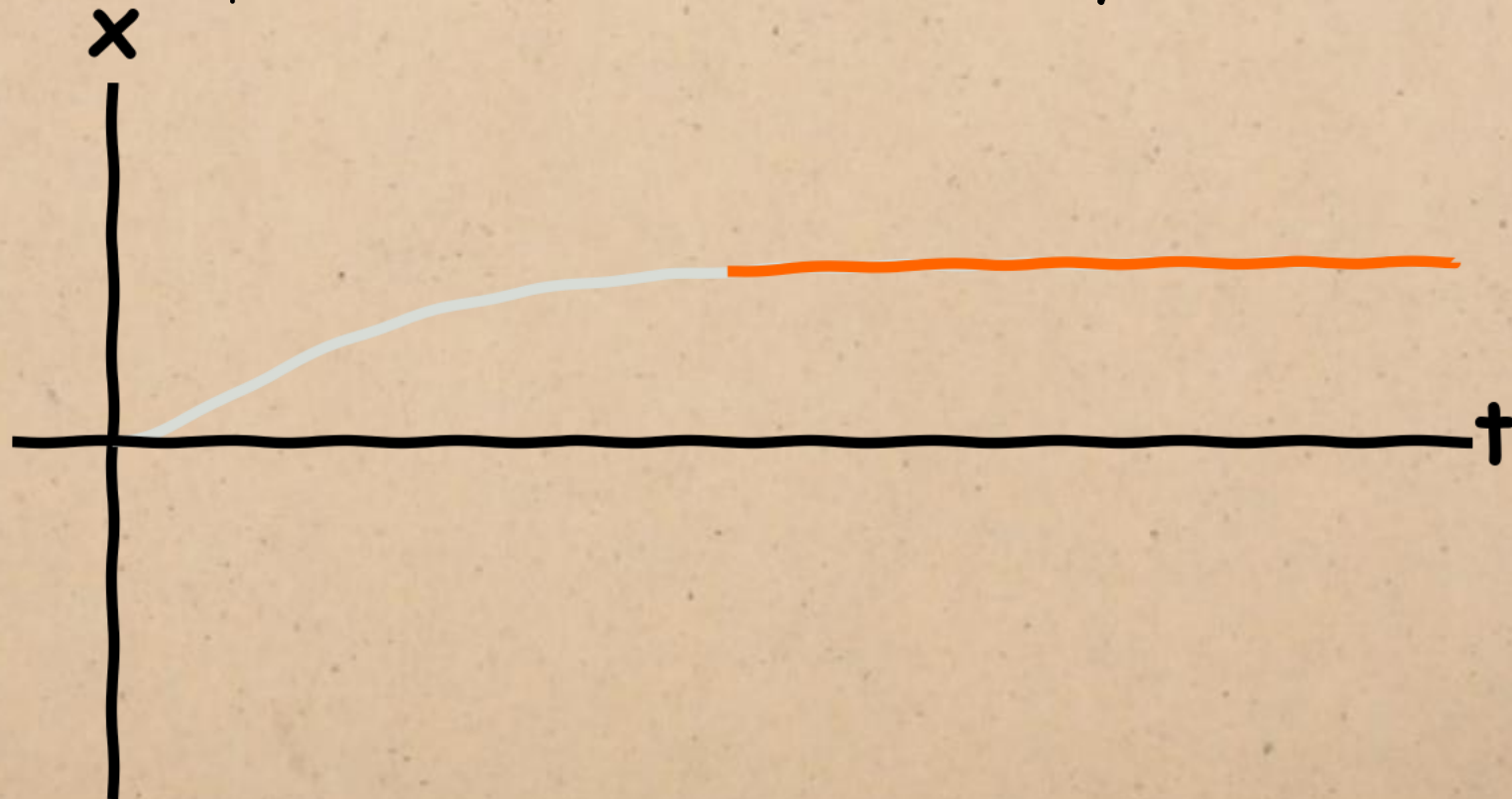
forced response = part of response
related to input

$$x(s) = \frac{1}{ms^2 + bs + k} F(s) + \frac{(b + ms)x(0) + m\dot{x}(0)}{ms^2 + bs + k}$$

free response = part of response
related to initial conditions

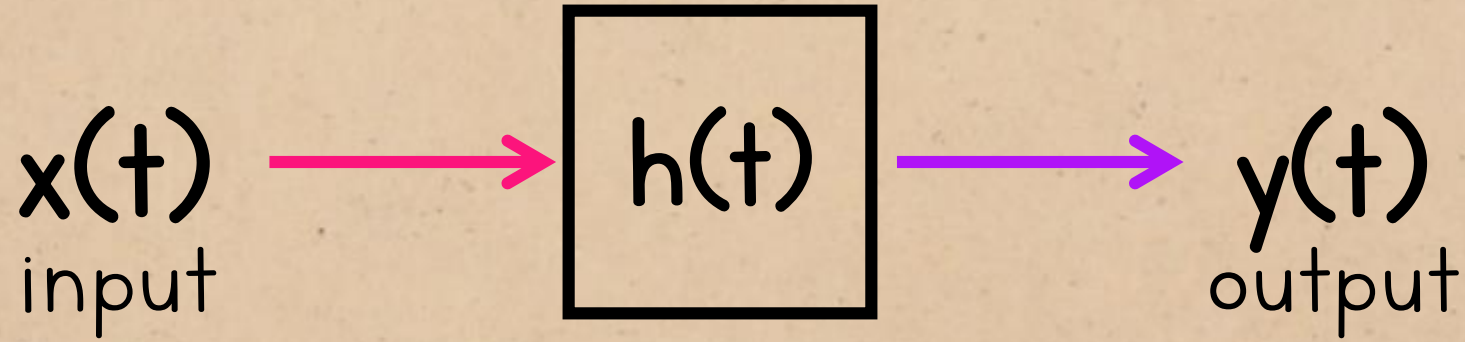
FREQUENCY RESPONSE

The frequency response is only a function of the input and concerns the **steady state**.



FREQUENCY RESPONSE

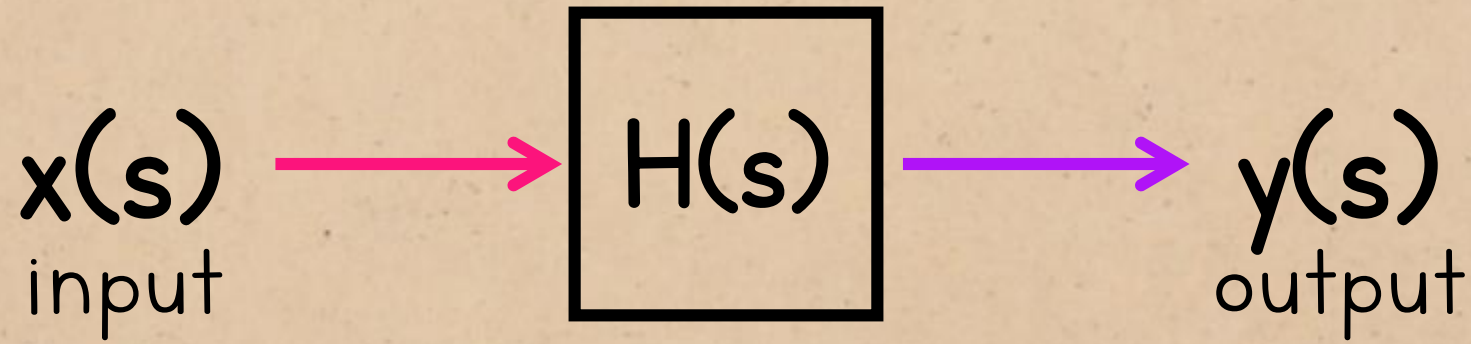
The **transfer function** is the Laplace transform of the impulse response.



$$y(t) = h(t) * x(t)$$

FREQUENCY RESPONSE

The **transfer function** is the Laplace transform of the impulse response.



$$y(s) = H(s) x(s)$$

FREQUENCY RESPONSE

Laplace synthesis equation:

$$h(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} H(s) e^{st} ds$$

(inverse Laplace transform)

FREQUENCY RESPONSE

Laplace synthesis equation:

$$h(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} H(s) e^{st} ds$$

The value of the transfer function at some value of gives the coefficient of e^{st} in the weighted sum that adds up to $h(t)$.

$$= \dots + \frac{1}{2\pi j} H(s) e^{st} + \dots$$

FREQUENCY RESPONSE

Laplace synthesis equation:

$$h(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} H(s) e^{st} ds$$

s is a complex variable: $s = \sigma + j\omega$

$$= \dots + \frac{1}{2\pi j} H(\sigma + j\omega) e^{\sigma t + j\omega t} + \dots$$

FREQUENCY RESPONSE

Laplace synthesis equation:

$$h(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} H(s) e^{st} ds$$

when we combine the complex conjugate pairs...

$$= \dots + \frac{1}{2\pi j} H(\sigma + j\omega) e^{\sigma t + j\omega t} + \frac{1}{2\pi j} H(\sigma - j\omega) e^{\sigma t - j\omega t} + \dots$$

FREQUENCY RESPONSE

Laplace synthesis equation:

$$h(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} H(s) e^{st} ds$$

when we combine the complex conjugate pairs...

$$= \dots + |H(\sigma + j\omega)| e^{\sigma t} \cos(j\omega t + \angle H(\sigma + j\omega)) + \dots$$

FREQUENCY RESPONSE

Laplace synthesis equation:

$$h(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} H(s) e^{st} ds$$

we get an exponential part associated with σ ...

$$= \dots + |H(\sigma + j\omega)| e^{\sigma t} \cos(j\omega t + \angle H(\sigma + j\omega)) + \dots$$

FREQUENCY RESPONSE

Laplace synthesis equation:

$$h(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} H(s) e^{st} ds$$

and a sinusoidal part associated with ω .

$$= \dots + |H(\sigma + j\omega)| e^{\sigma t} \cos(j\omega t + \angle H(\sigma + j\omega)) + \dots$$

FREQUENCY RESPONSE

Laplace synthesis equation:

$$h(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} H(s) e^{st} ds$$



If the system is stable, $e^{\sigma t}$ will die away over time...

$$= \dots + |H(\sigma + j\omega)| e^{\sigma t} \cos(j\omega + \angle H(\sigma + j\omega)) + \dots$$

FREQUENCY RESPONSE

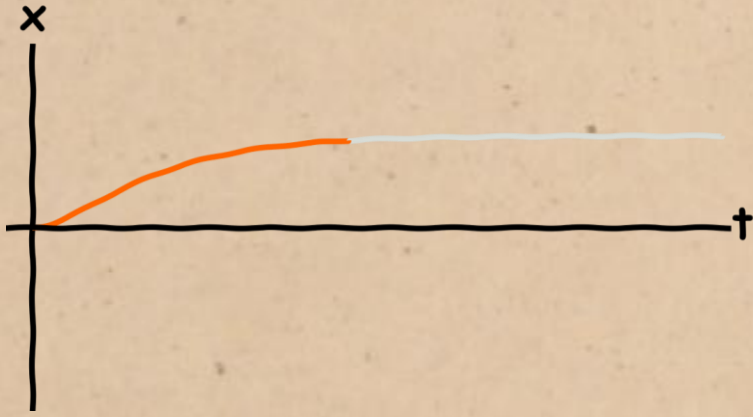
Laplace synthesis equation:

$$h(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} H(s) e^{st} ds$$

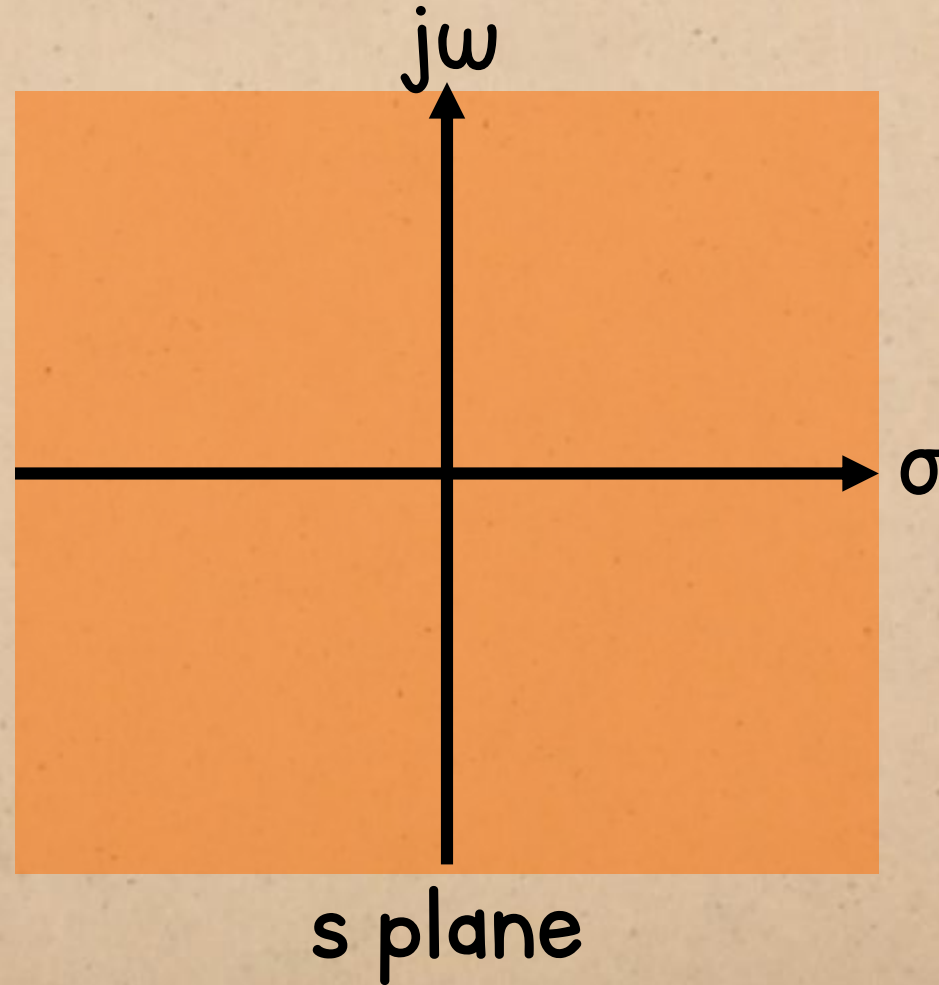
and we'll be left with only the components where $\sigma = 0$

$$= \dots + |H(j\omega)| \cos(j\omega + \angle H(j\omega)) + \dots$$

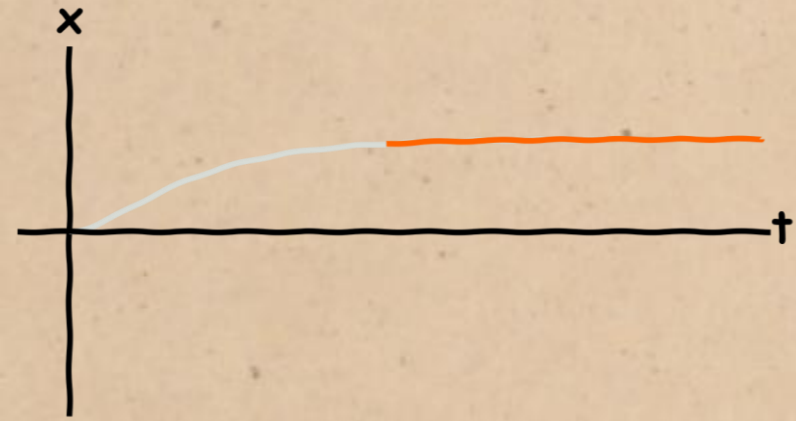
FREQUENCY RESPONSE



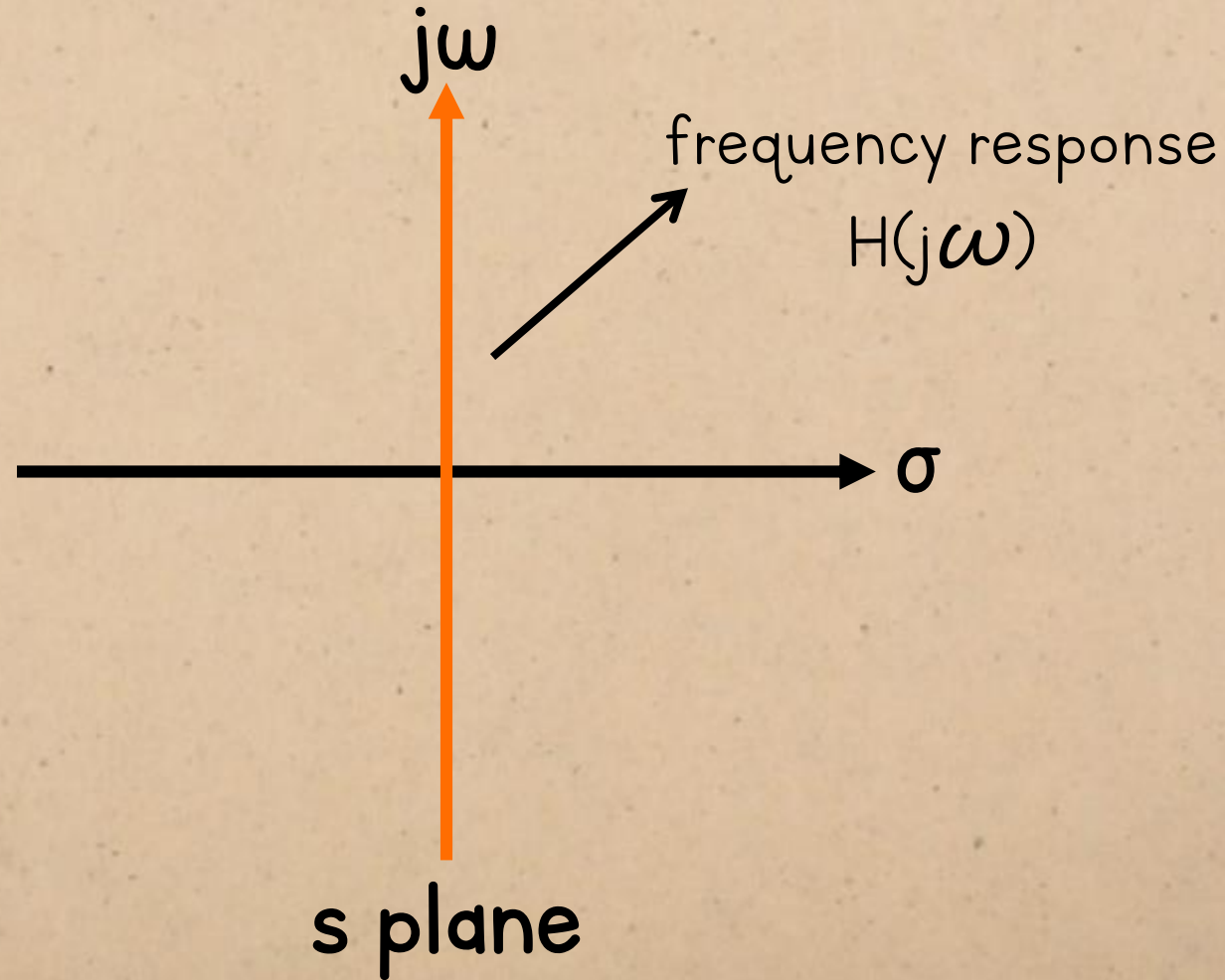
The components from s values with nonzero real parts make up the transient response.



FREQUENCY RESPONSE



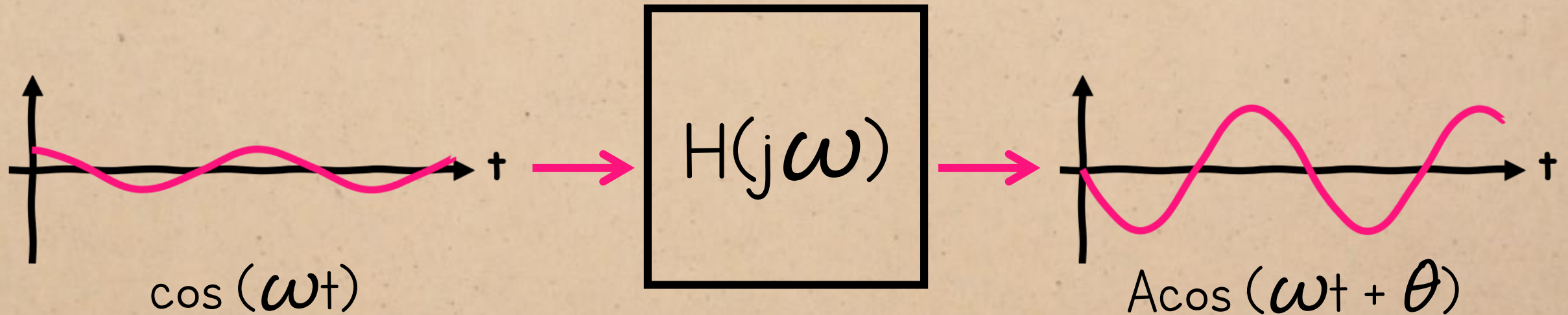
The s values along the imaginary axis contribute to the **steady state**.



FREQUENCY RESPONSE

The frequency response value for some $j\omega$ will be a complex value:

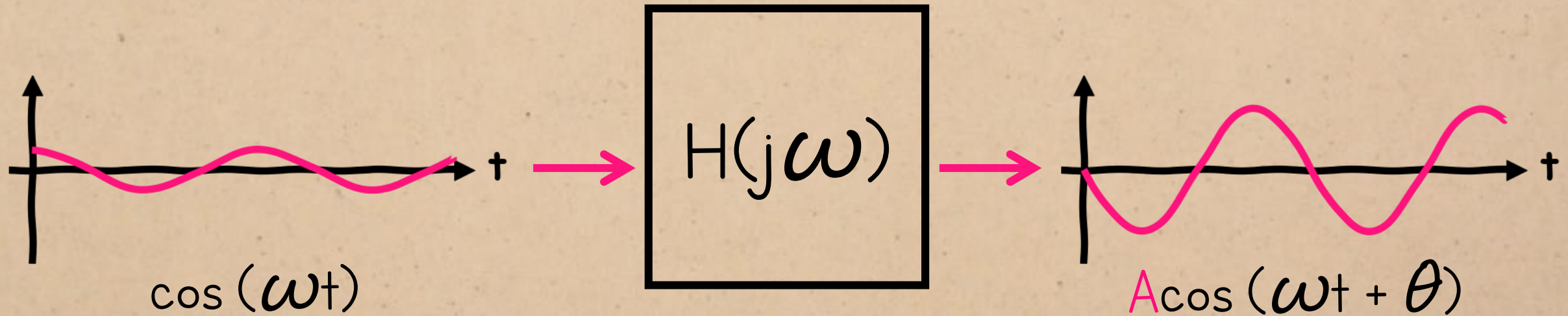
$$H(j\omega) = Ae^{j\theta}$$



FREQUENCY RESPONSE

The frequency response value for some $j\omega$ will be a complex value:

$$H(j\omega) = Ae^{j\theta}$$

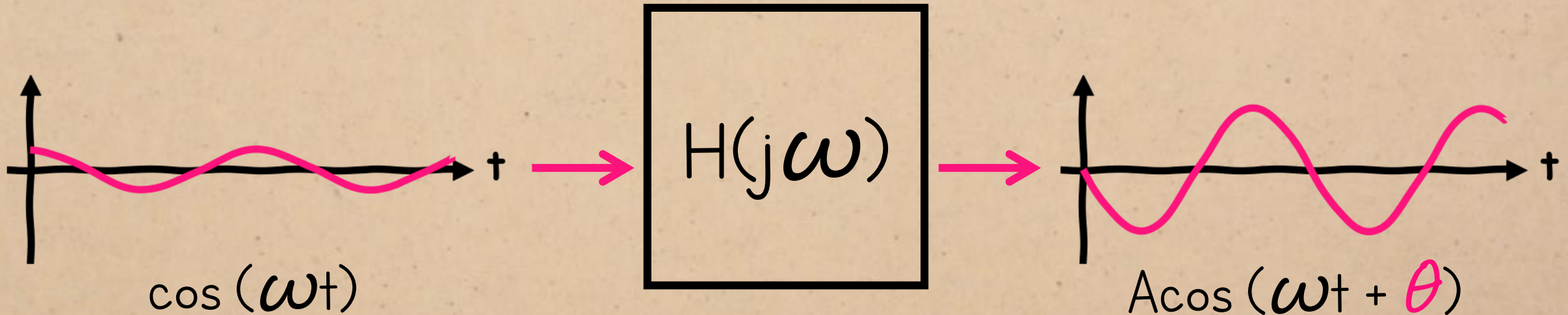


real part = the **gain** at
that frequency

FREQUENCY RESPONSE

The frequency response value for some $j\omega$ will be a complex value:

$$H(j\omega) = Ae^{j\theta}$$

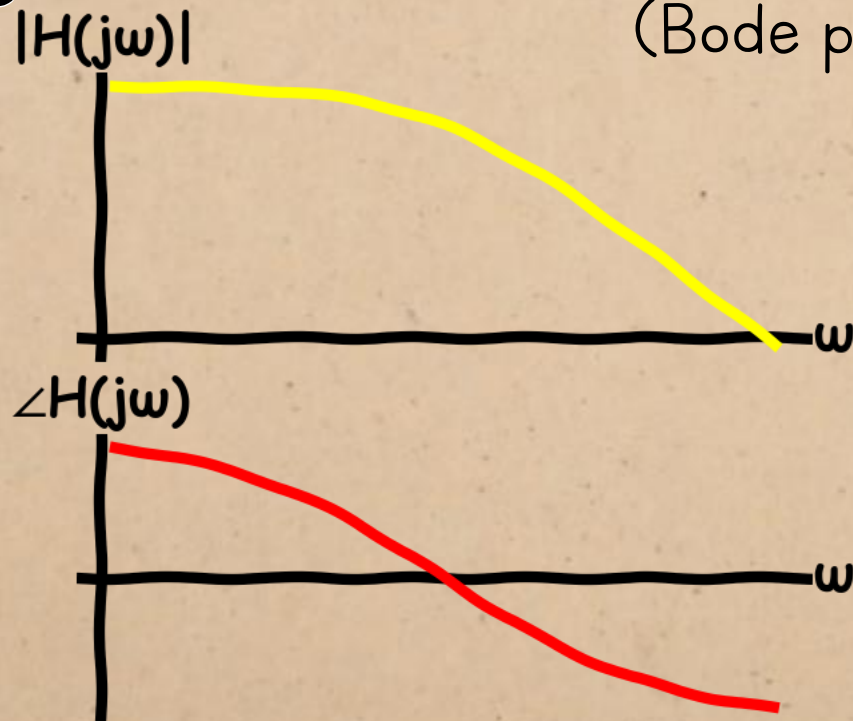


imaginary part =
the **phase shift**

FREQUENCY RESPONSE

There are two ways we can represent the frequency response graphically:

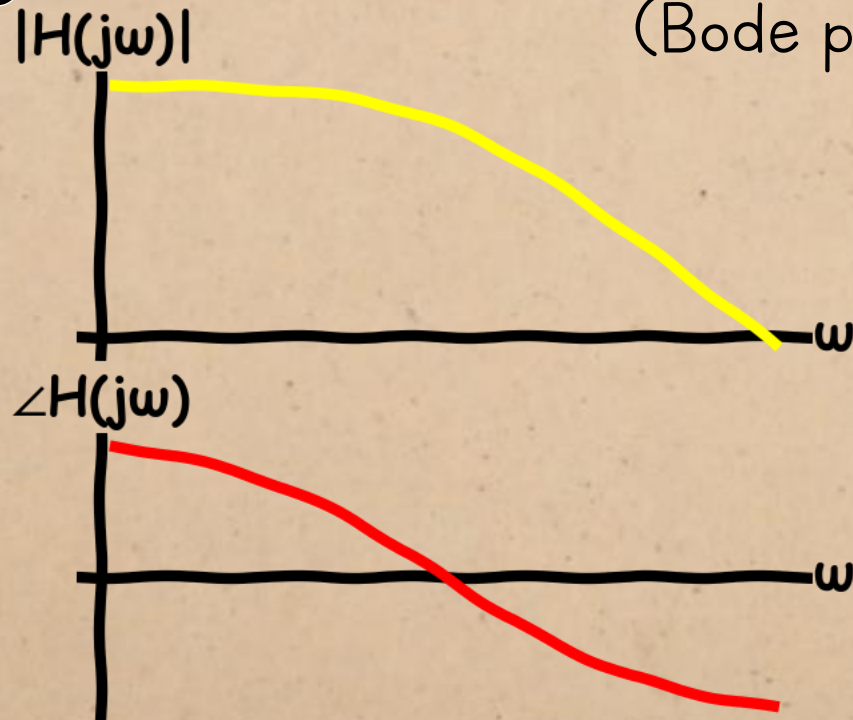
- 1. separate magnitude and phase plots:
(Bode plot)



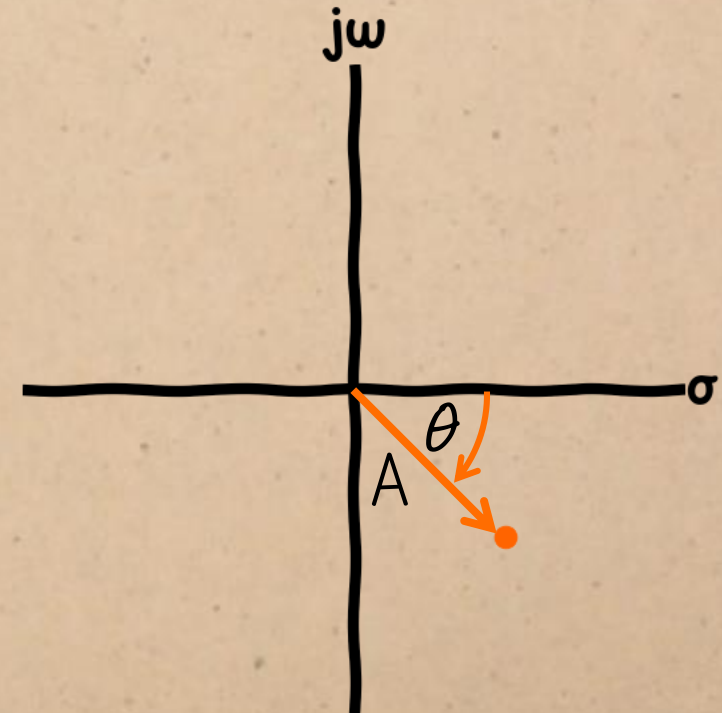
FREQUENCY RESPONSE

There are two ways we can represent the frequency response graphically:

1. separate magnitude and phase plots:
(Bode plot)



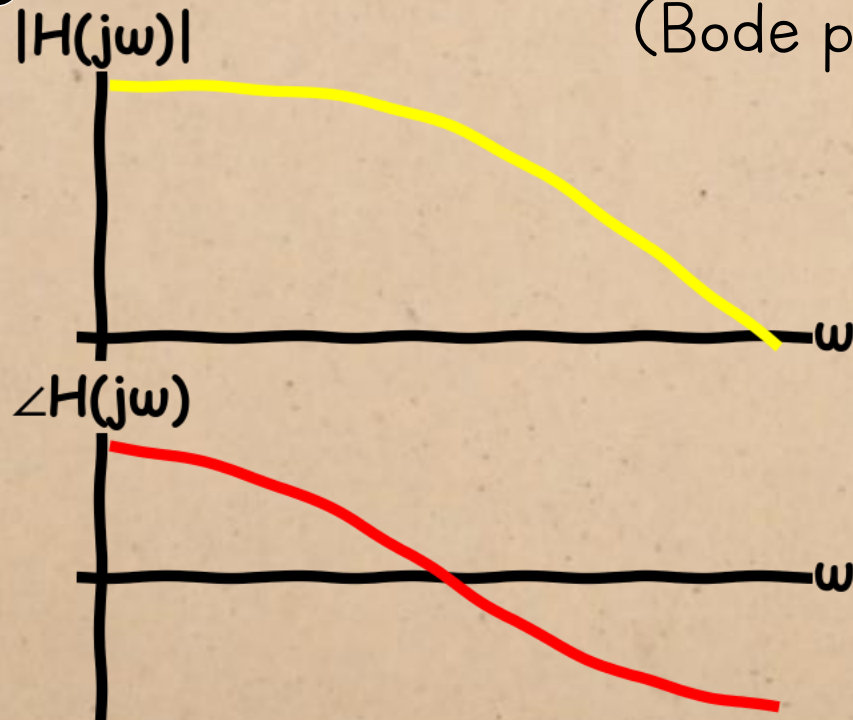
2. magnitude and phase on same set of axes:



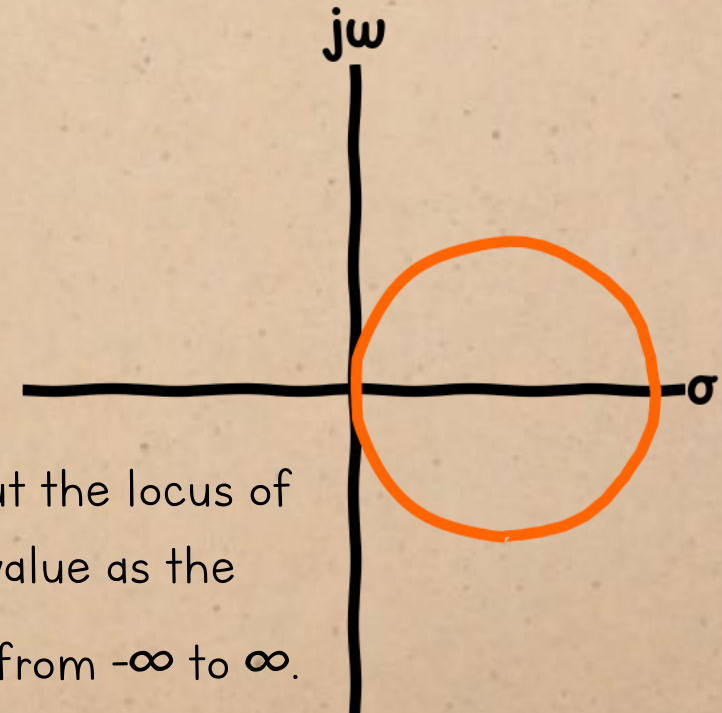
FREQUENCY RESPONSE

There are two ways we can represent the frequency response graphically:

1. separate magnitude and phase plots:
(Bode plot)



2. magnitude and phase on same set of axes:

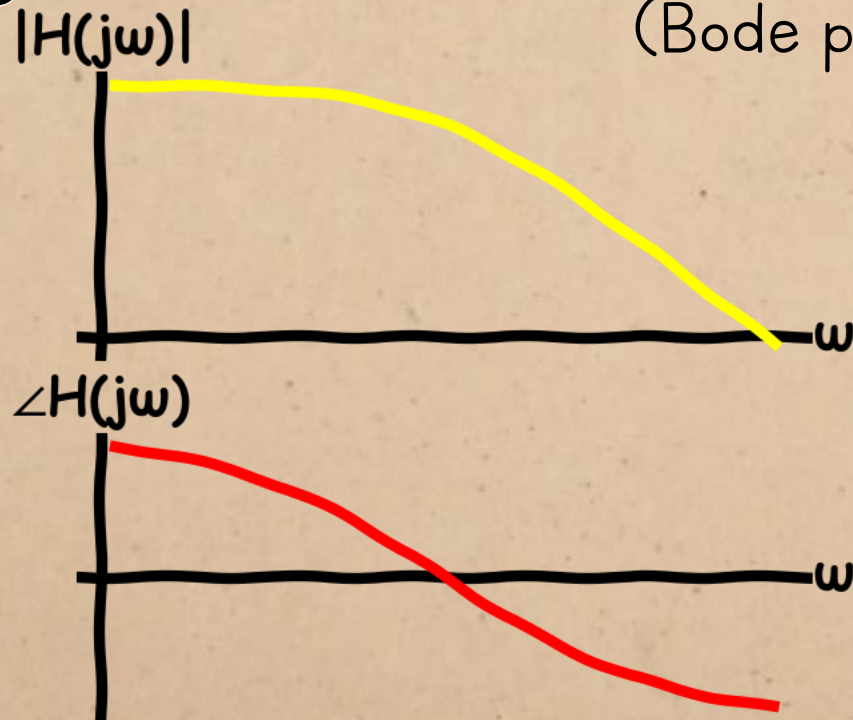


We then trace out the locus of the response value as the frequency varies from $-\infty$ to ∞ .

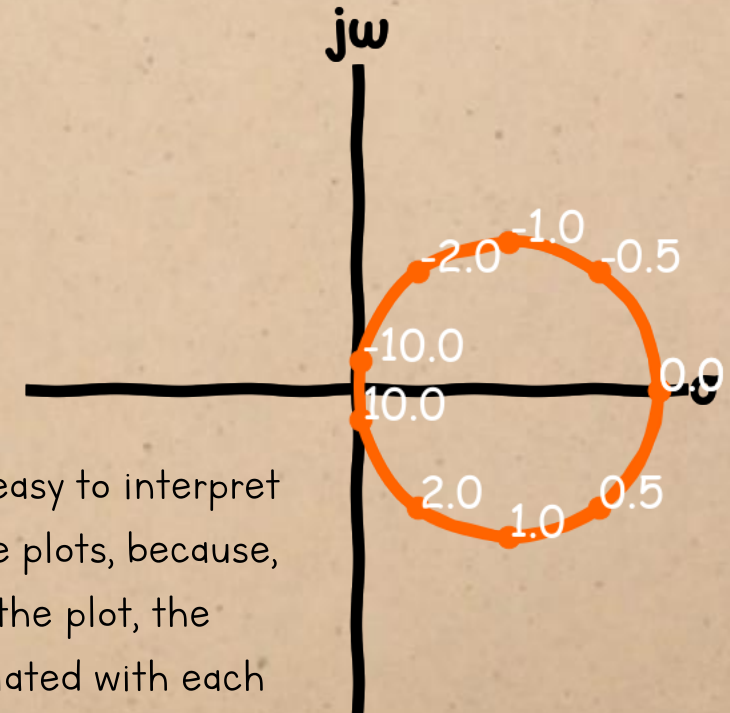
FREQUENCY RESPONSE

There are two ways we can represent the frequency response graphically:

1. separate magnitude and phase plots:
(Bode plot)



2. magnitude and phase on same set of axes:

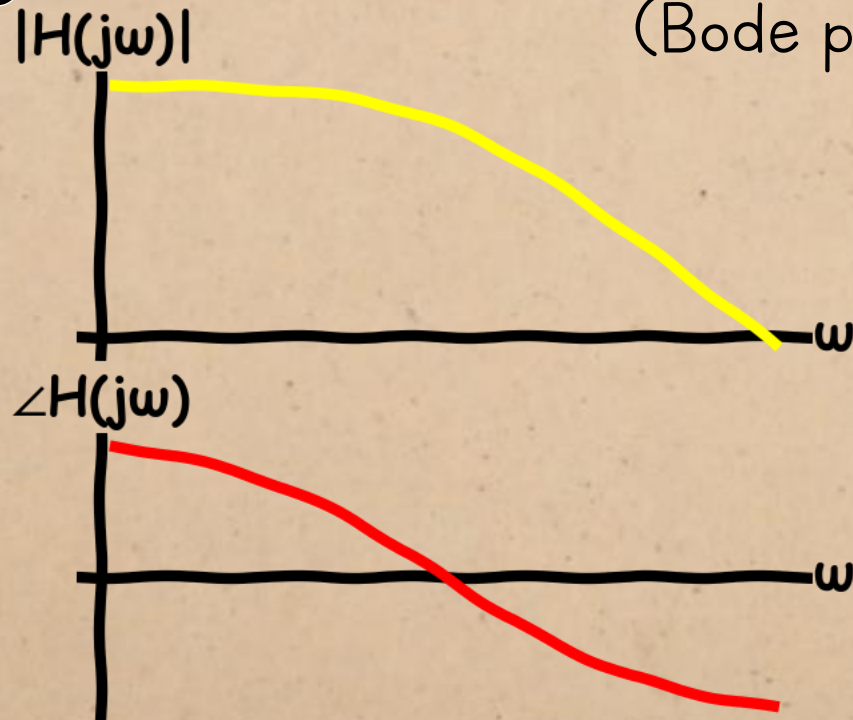


This approach is not as easy to interpret as the one with separate plots, because, unless we annotate the plot, the frequency values associated with each point aren't immediately visible.

FREQUENCY RESPONSE

There are two ways we can represent the frequency response graphically:

1. separate magnitude and phase plots:
(Bode plot)



2. magnitude and phase on same set of axes:
(Nyquist plot)

...But it does have its own advantages, which we'll see when we look at the Nyquist plot.

